

Lecture 06

Tradable Permits

Ivan Rudik

AEM 4510

Roadmap

1. How do tradable permit systems work in theory and in the real world?
2. What happens under a tradable permit system?

Tradable permits

Tradable permits

How do tradable permit systems work?¹

First, recall a regular emission standard: we set $\bar{E}$ at the point where $MAC = MD$

This is easy with one firm, but what if we have several, or hundreds?

¹ Tradable permit systems are also called cap and trade systems.

Optimal policy with multiple firms

Suppose the regulator chooses emissions for N firms: $E_1, \dots, E_N$

Let firm i 's abatement cost be $C_i(E_i)$

Let total damages depend on aggregate emissions:

$$D = D\left(\sum_{i=1}^N E_i\right)$$

Regulator problem:

$$\min_{E_1, \dots, E_N} \sum_{i=1}^N C_i(E_i) + D\left(\sum_{i=1}^N E_i\right)$$

Optimal policy with multiple firms

Take one FOC for each firm $i = 1, \dots, N$:

$$\frac{\partial}{\partial E_i} \left[\sum_{j=1}^N C_j(E_j) + D \left(\sum_{j=1}^N E_j \right) \right] = 0$$

$$C'_i(E_i) + D' \left(\sum_{j=1}^N E_j \right) = 0$$

$$\underbrace{-C'_i(E_i)}_{MAC_i} = \underbrace{D' \left(\sum_{j=1}^N E_j \right)}_{MD}$$

Optimal policy with multiple firms

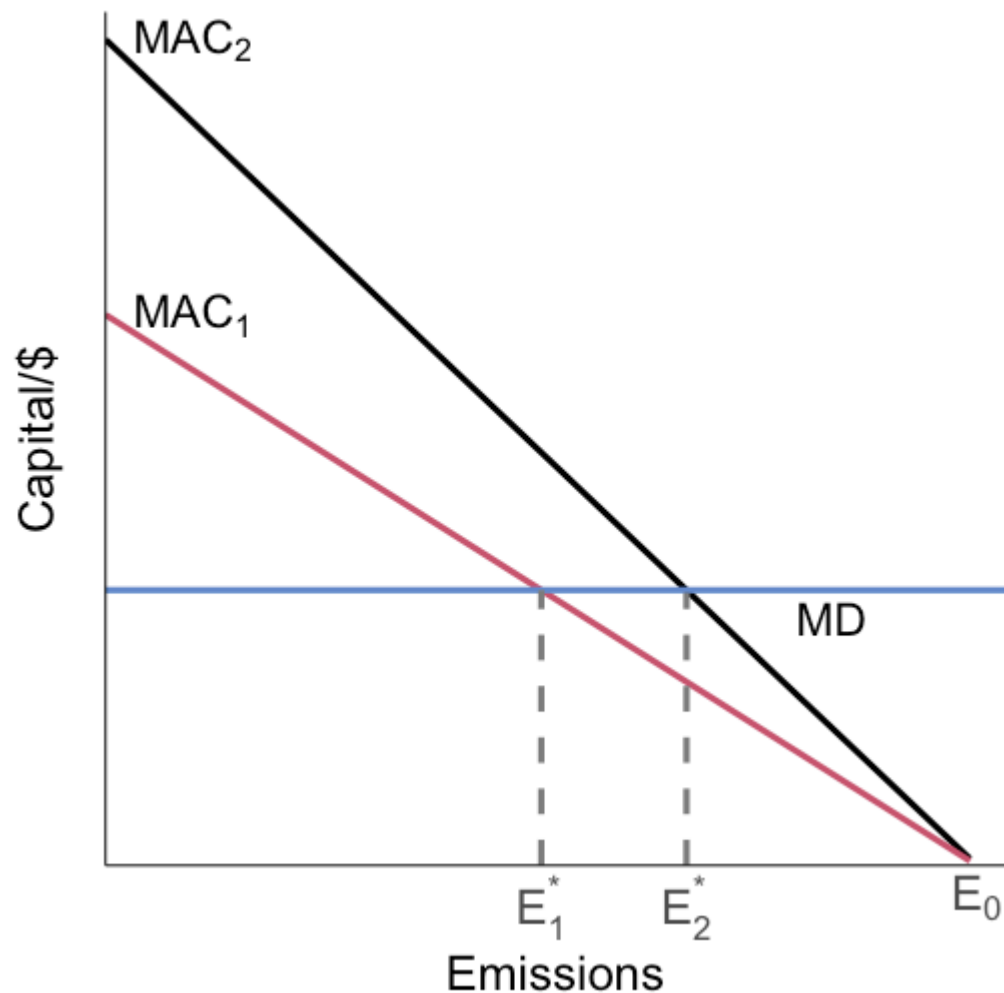
$$\underbrace{-C'_i(E_i)}_{MAC_i} = \underbrace{D' \left(\sum_{j=1}^N E_j \right)}_{MD}$$

At the optimum:

$$MAC_1 = MAC_2 = \dots = MAC_N = MD$$

This gives the target condition that we try to achieve through a decentralized permit market

Optimal policy with multiple firms



Firm #2 is 'dirty': has higher MAC

Firm #1 is 'clean': has lower MAC

If we use a regular emission standard: it has to be firm-specific!

Mandate E_1^* for 1 and E_2^* for 2

This requires **a lot** of info and political capital on behalf of the regulator

Tradable permit systems

So how do these systems make standards more flexible?

They allow firms to trade their emission allowances

E.g. if firms are restricted to $\bar{E}_1$ and $\bar{E}_2$, we can allow the firms to trade

If firm 1 sells an allowance/permit to firm 2, their new restrictions are: $\bar{E}_1 - 1$ and $\bar{E}_2 + 1$

Total emissions didn't change, but how much each firm gets to emit does through voluntary trade

Tradable permit systems: example

The US Acid Rain Program is the classic example

Permit = license to create 1 ton of SO₂

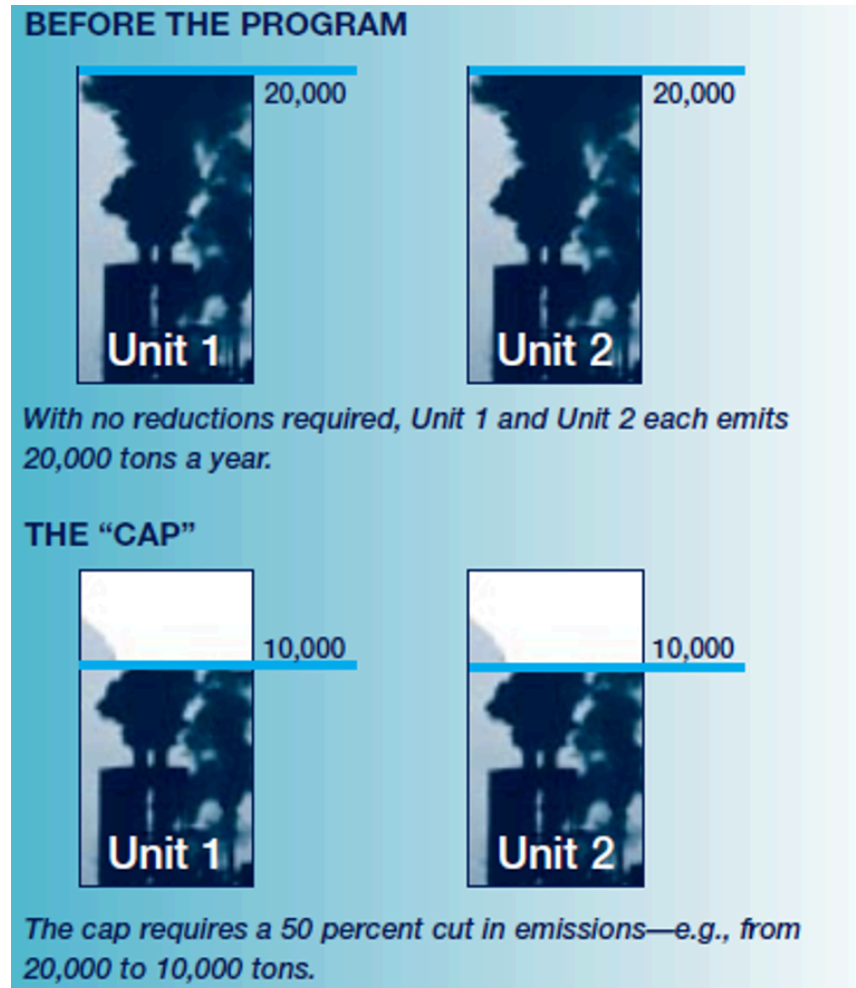
Phase I (1995-2000):

- 6.3 million permits issued per year
- affected 263 generating units at 110 dirtiest power plants

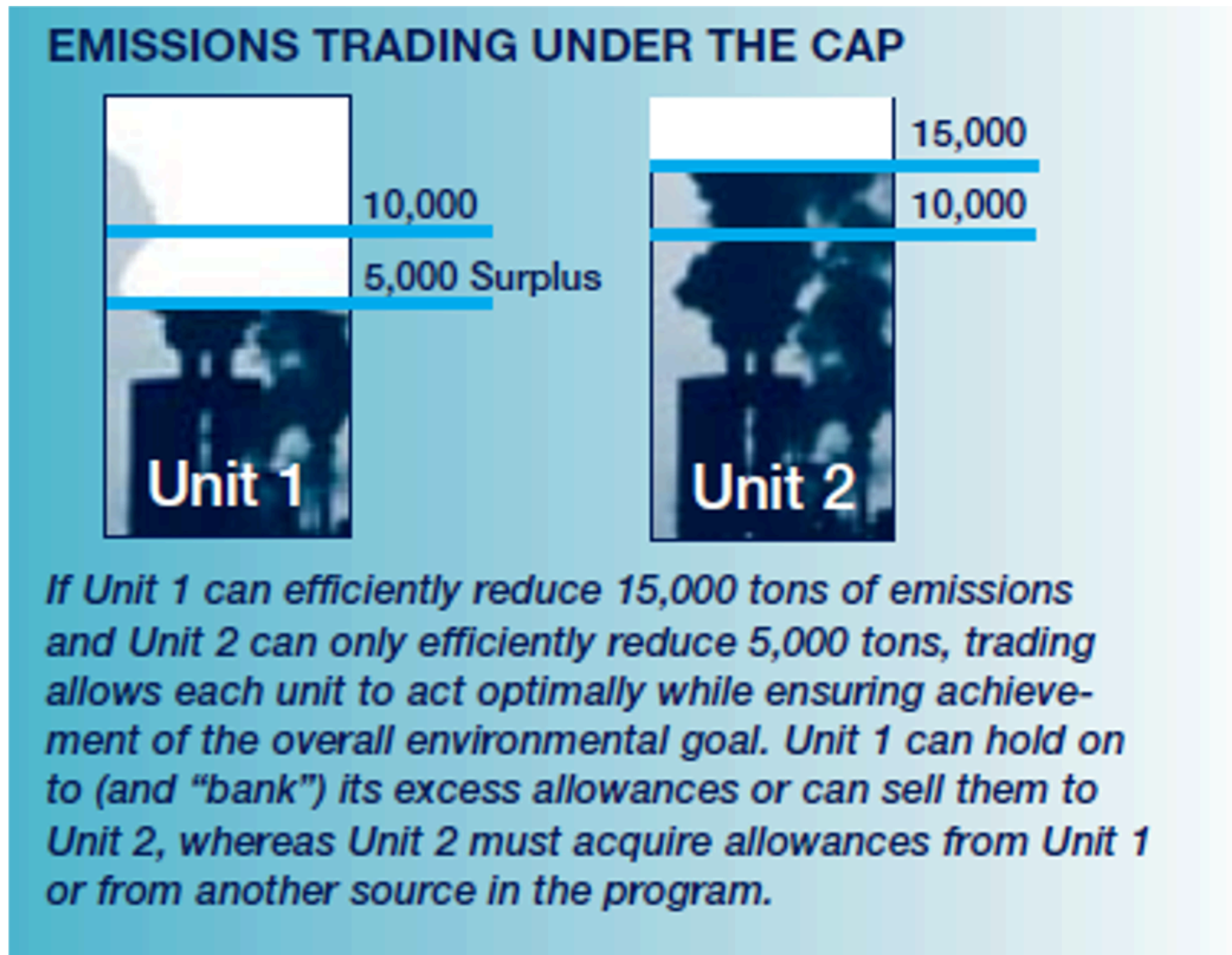
Phase II (2000+):

- 9 million permits issued per year
- affects all power plants over some minimum size

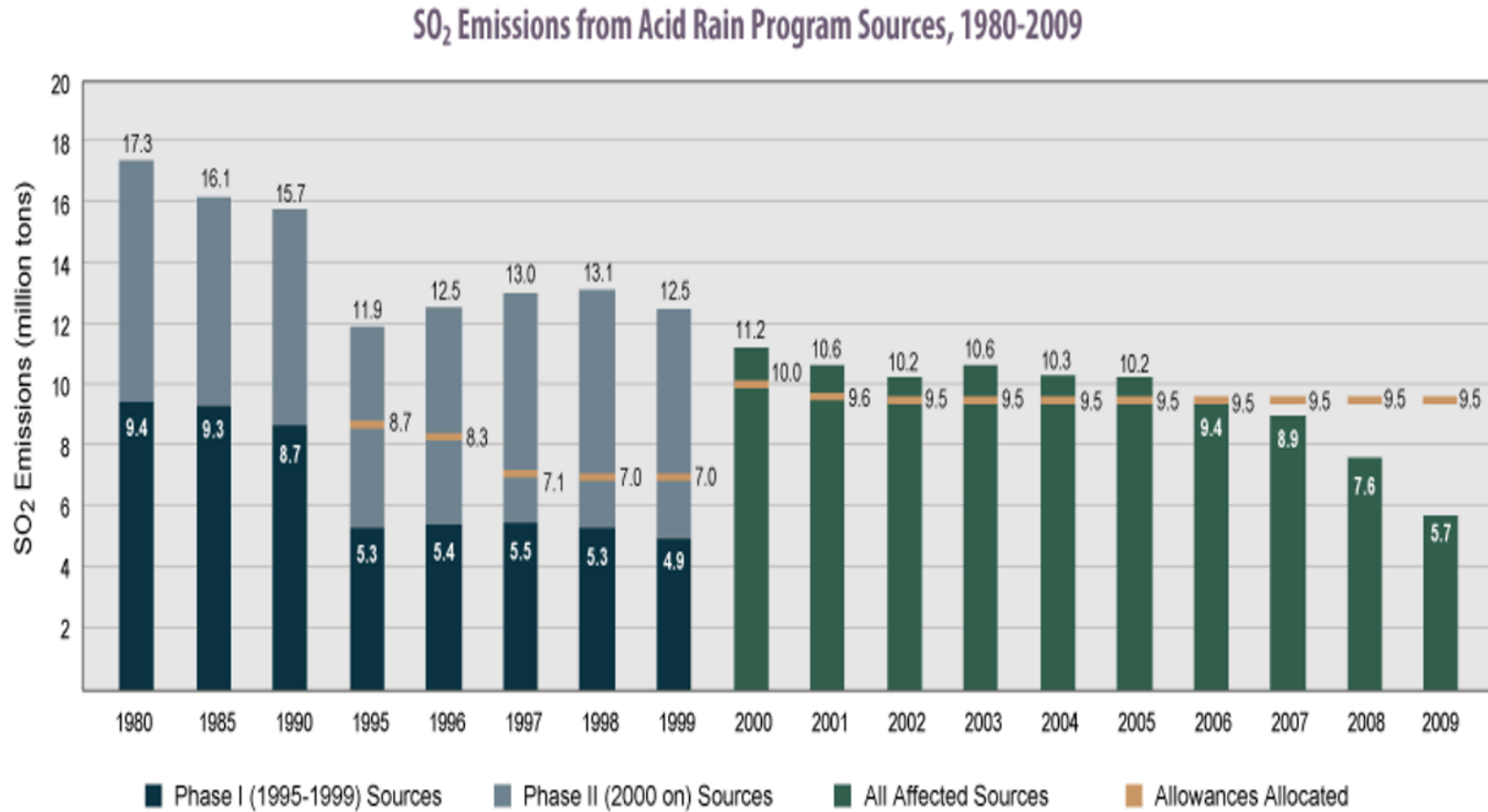
Tradable permit systems: example



Tradable permit systems: example

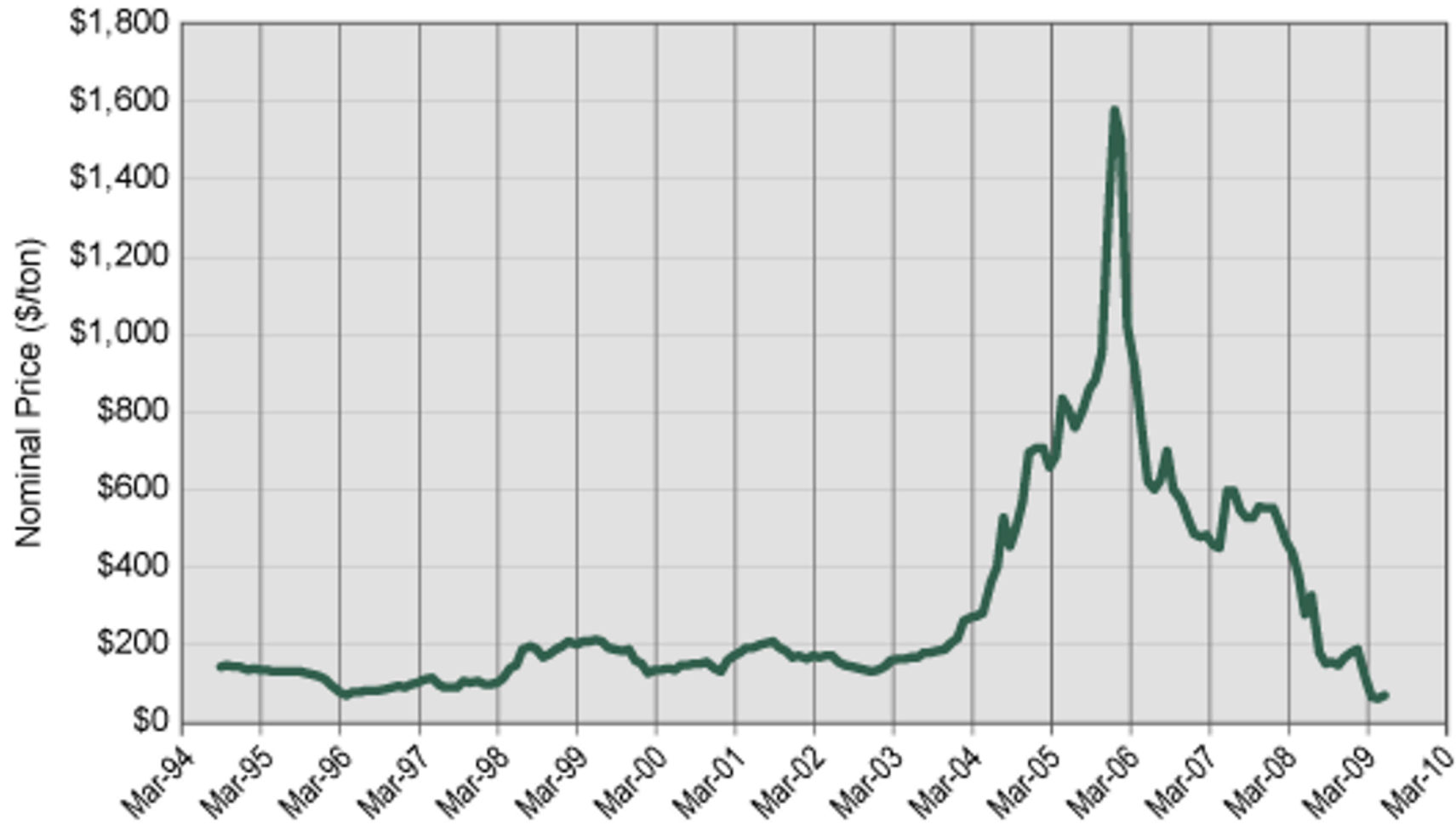


Tradable permit systems: example



Source: EPA, 2010

Tradable permit systems: example



Tradable permit systems: example

Quantified benefits*:

PM_{2.5} mortality (U.S. and southern Canada)	\$107,000
PM_{2.5} morbidity (U.S. and southern Canada)	\$8,000
Ozone mortality (eastern U.S.)	\$4,000
Ozone morbidity (eastern U.S.)	\$300
Visibility at parks (3 U.S. regions)	\$2,000
Recreational fishing in NY	\$65
Ecosystem improvements in Adirondacks (NY residents)	\$500

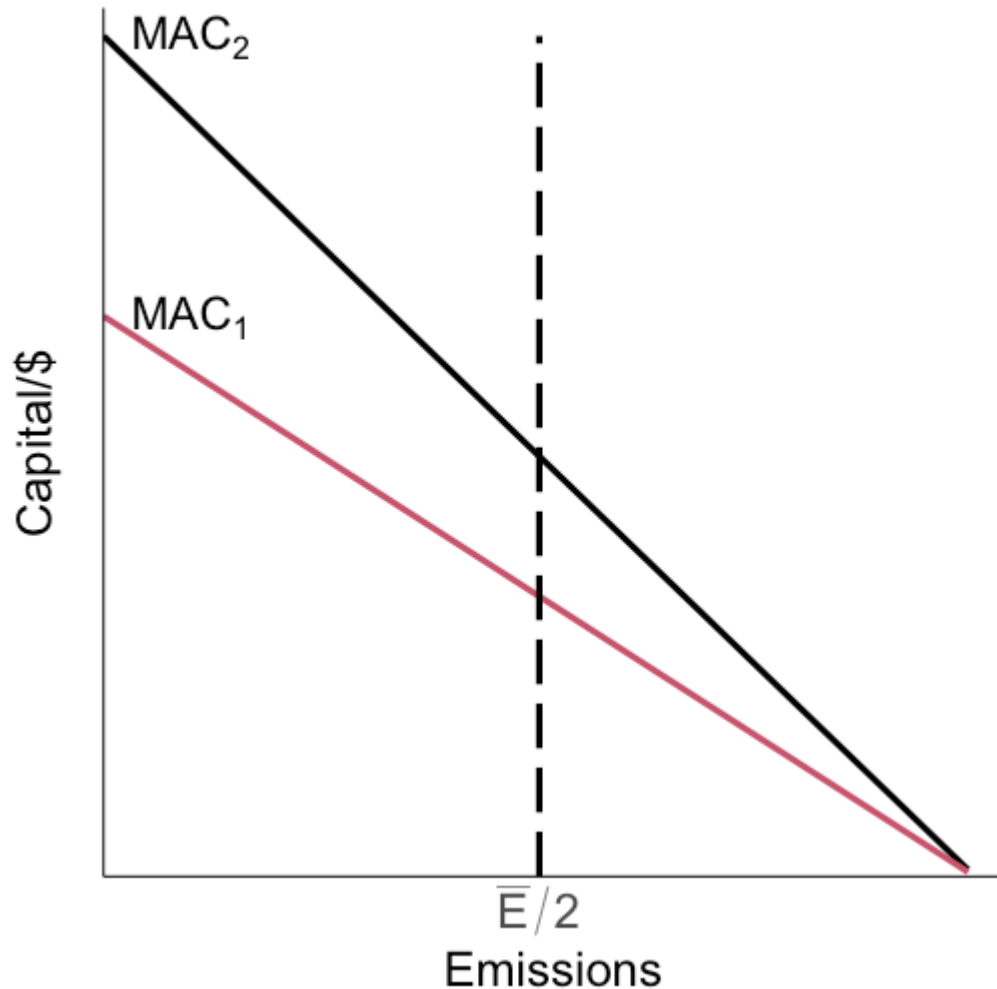
Total annual quantified benefits	\$122,000
---	------------------

Quantified costs for U.S. power generation:

SO₂ controls	\$2,000
NO_x controls	\$1,000

Total annual quantified costs	\$3,000
--------------------------------------	----------------

Tradable permits: graphical



Suppose we want to limit to $\bar{E}$ total emissions so each firm gets $\bar{E}/2$ permits, but can't trade them

This can't be efficient (i.e. maximize social welfare given some MD curve)

It also can't be cost-effective: it doesn't minimize the cost of achieving $\bar{E}$ total emissions

Tradable permits: cost-effectiveness

For cost-effectiveness, we need total costs to be minimized for achieving a given level of emissions:

$$\min_{E_1, E_2} C_1(E_1) + C_2(E_2) \quad \text{subject to: } E_1 + E_2 = \bar{E}$$

This is the same problem as:

$$\min_{E_1, E_2} C_1(E_1) + C_2(\bar{E} - E_1)$$

which has a solution where:

$$-C'_1(E_1^*) = -C'_2(\bar{E} - E_1^*)$$

Tradable permits: cost-effectiveness

Cost-effectiveness requires:

$$-C'_1(E_1^*) = -C'_2(\bar{E} - E_1^*) \leftrightarrow MAC_1 = MAC_2$$

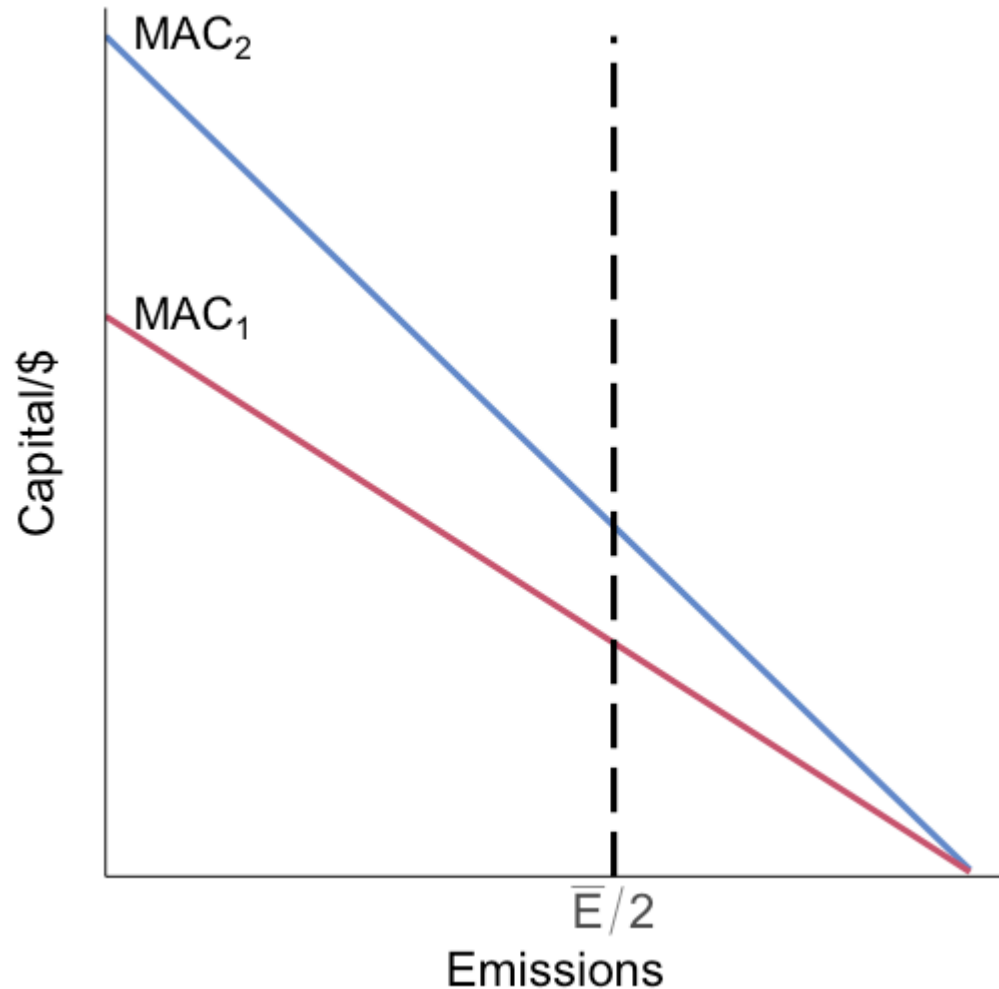
That marginal abatement costs are equal across all emitters

If firms have different MAC curves, giving them the same amount of emissions/permits cannot be optimal

How do we fix this?

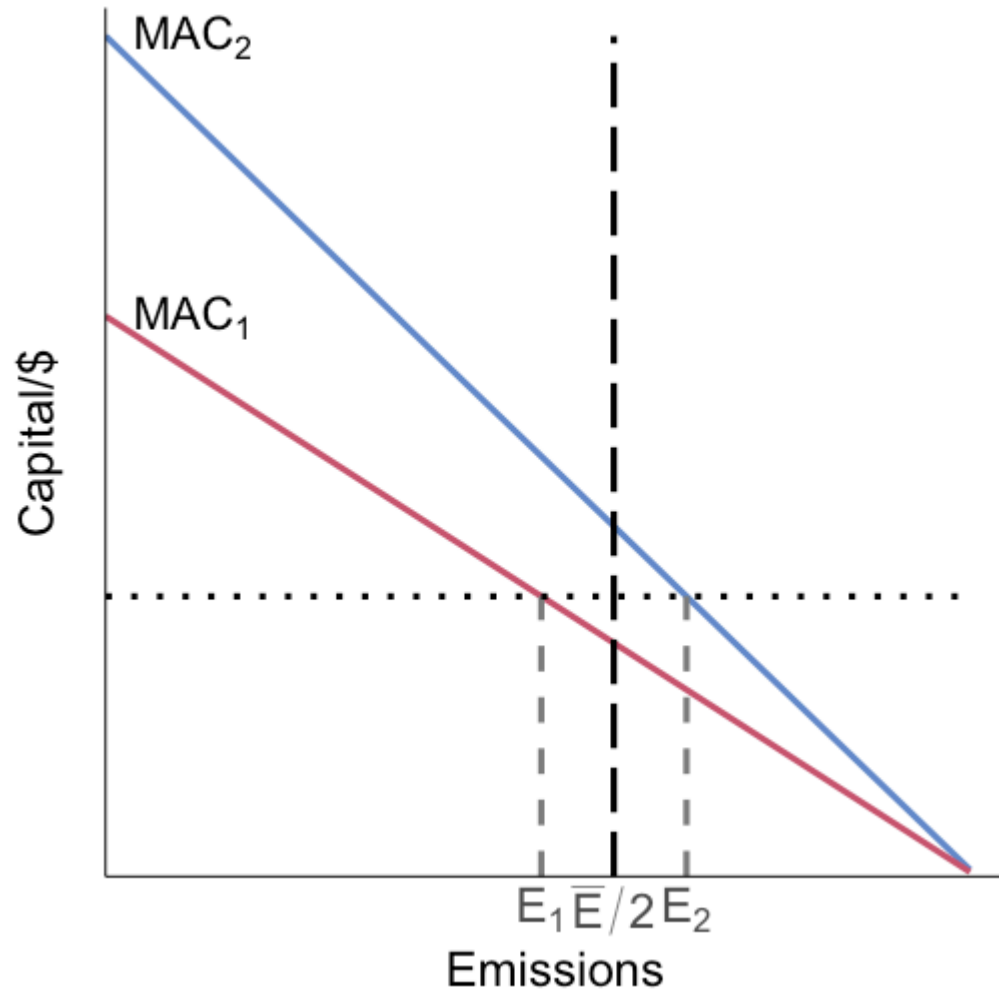
Let them trade the permits

Tradable permits: graphical



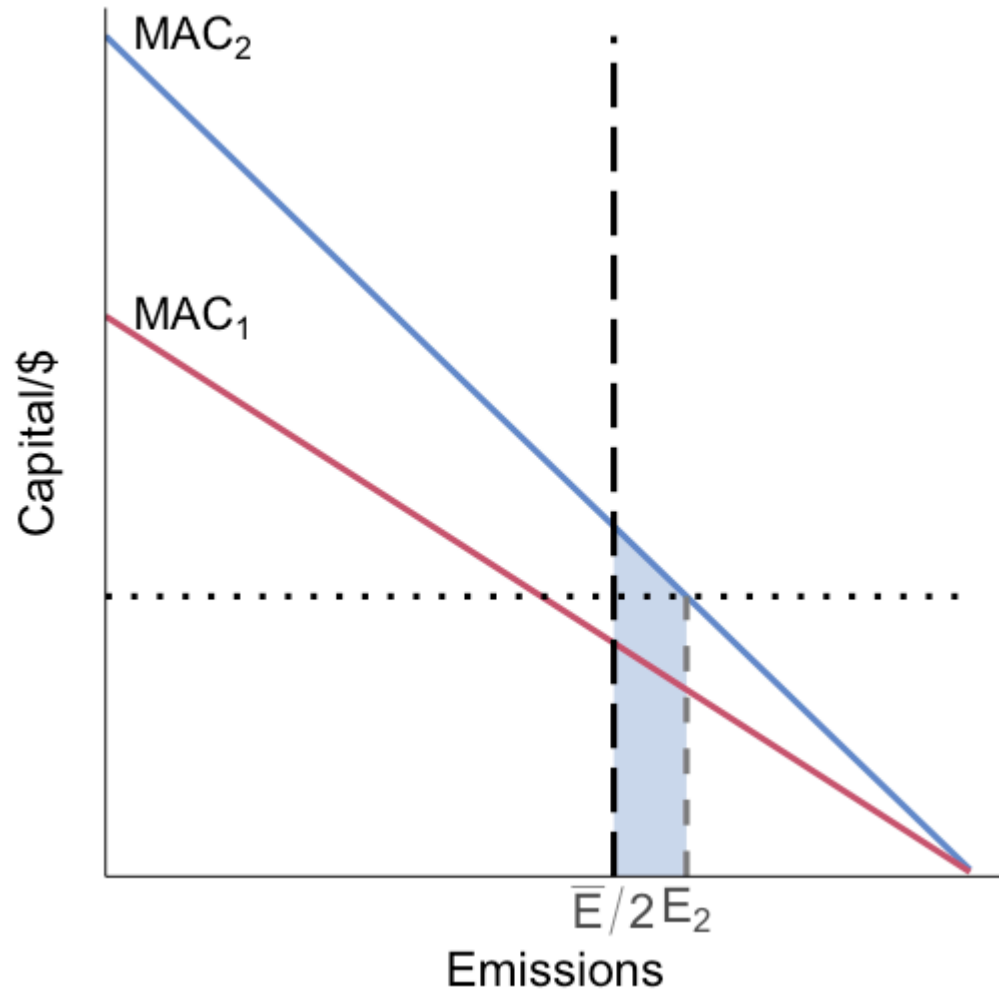
We can reduce costs by increasing abatement at which firm, and decreasing abatement at which firm?

Tradable permits: graphical



We can reduce costs by increasing emissions at high MAC firm 2 and decreasing emissions at low MAC firm 1 until they are equal

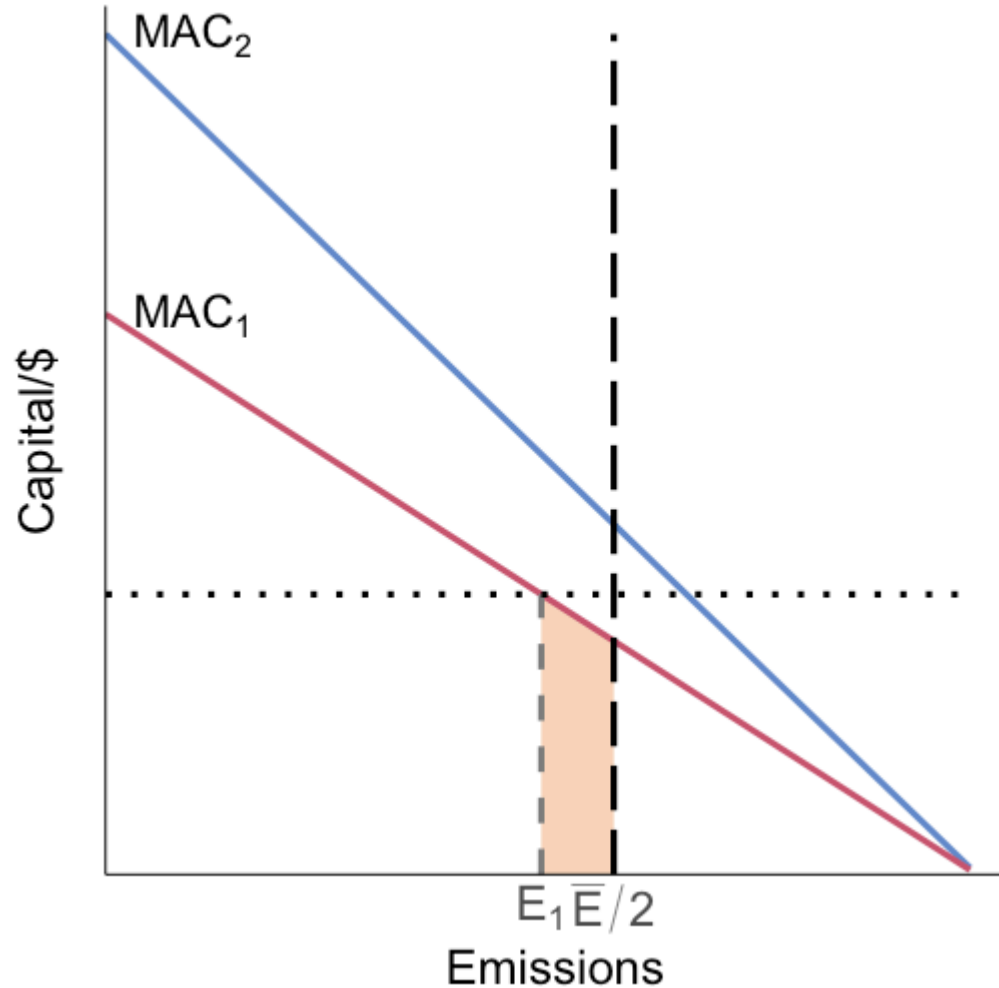
Tradable permits: graphical



Step 1: allow firm 2 (high MAC) to emit more, from $\bar{E}/2$ to E_2

The blue shaded area is firm 2's cost saving

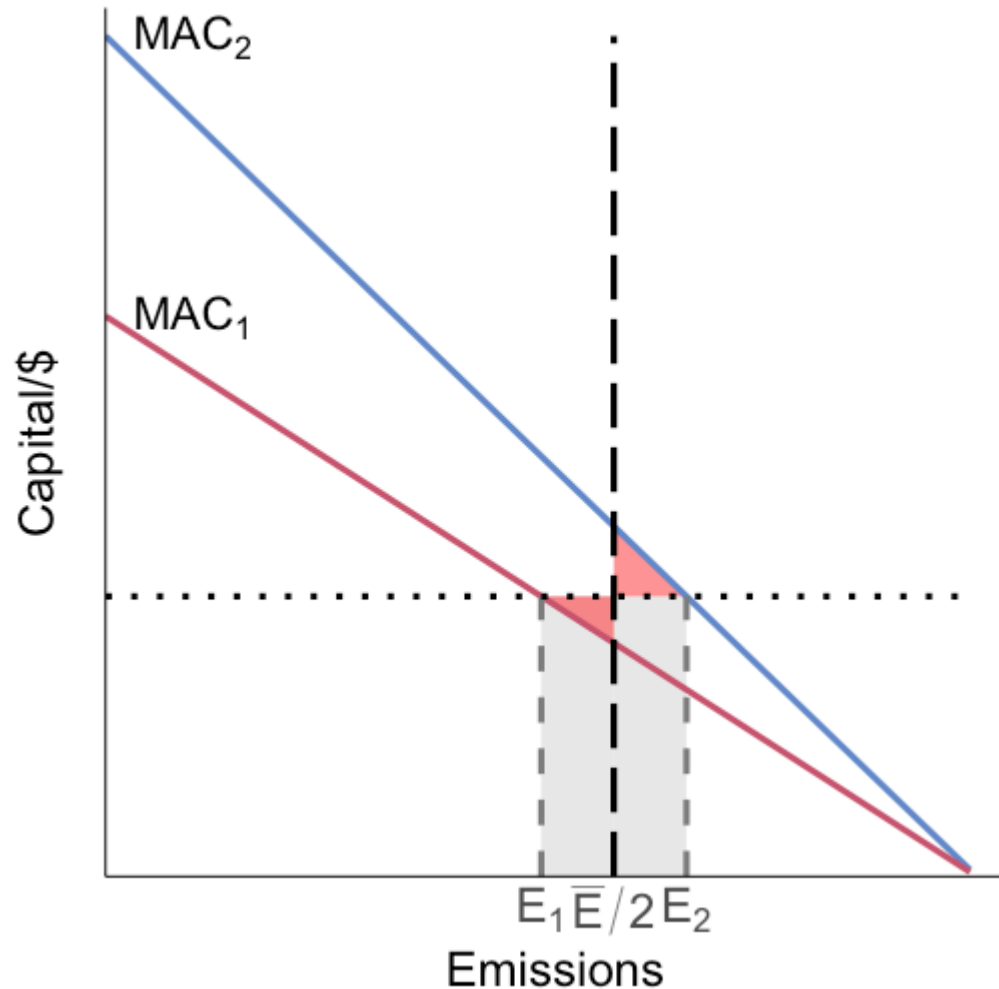
Tradable permits: graphical



Step 2: firm 1 (low MAC) emits less, from $\bar{E}/2$ to E_1

The orange shaded area is firm 1's cost increase

Tradable permits: graphical

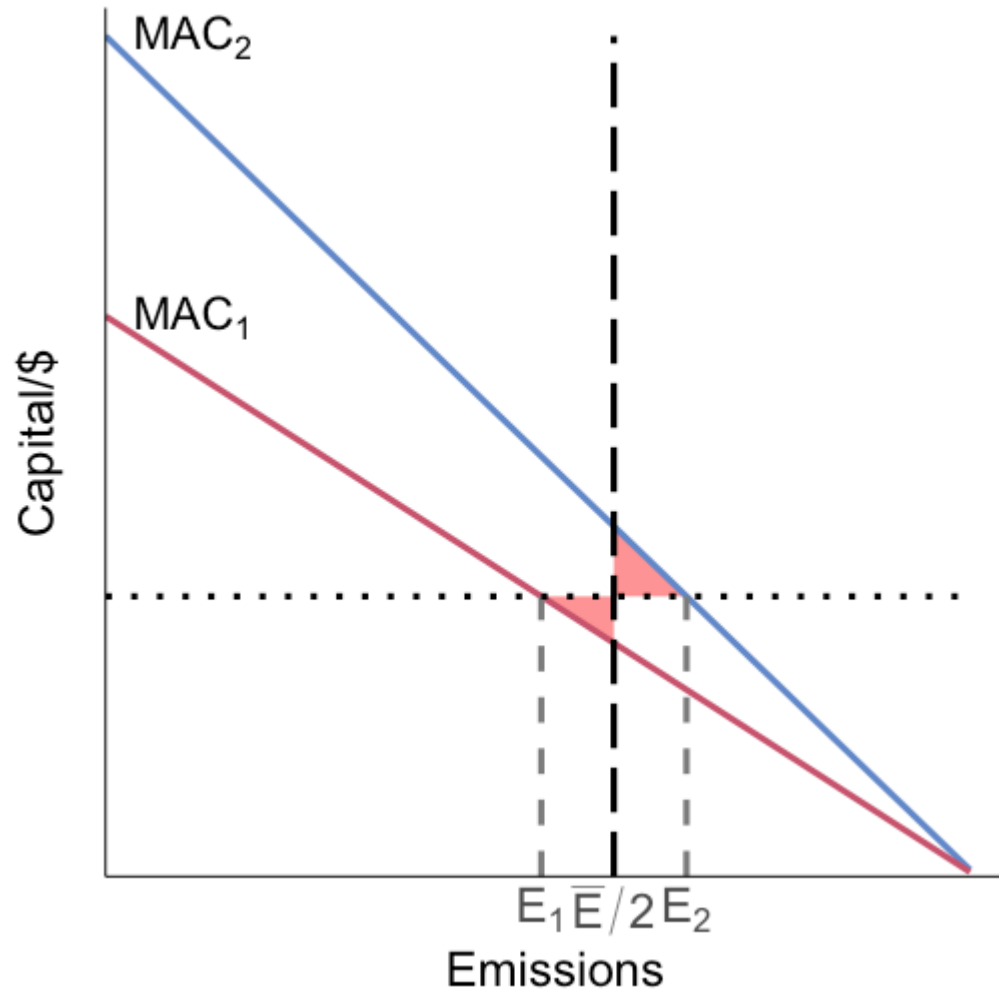


Step 3: net cost change from trading

The grey rectangles are equal (same height and same width ΔE), so they cancel out

What remains is the two red triangles: the net cost reduction from reallocation

Tradable permits: graphical

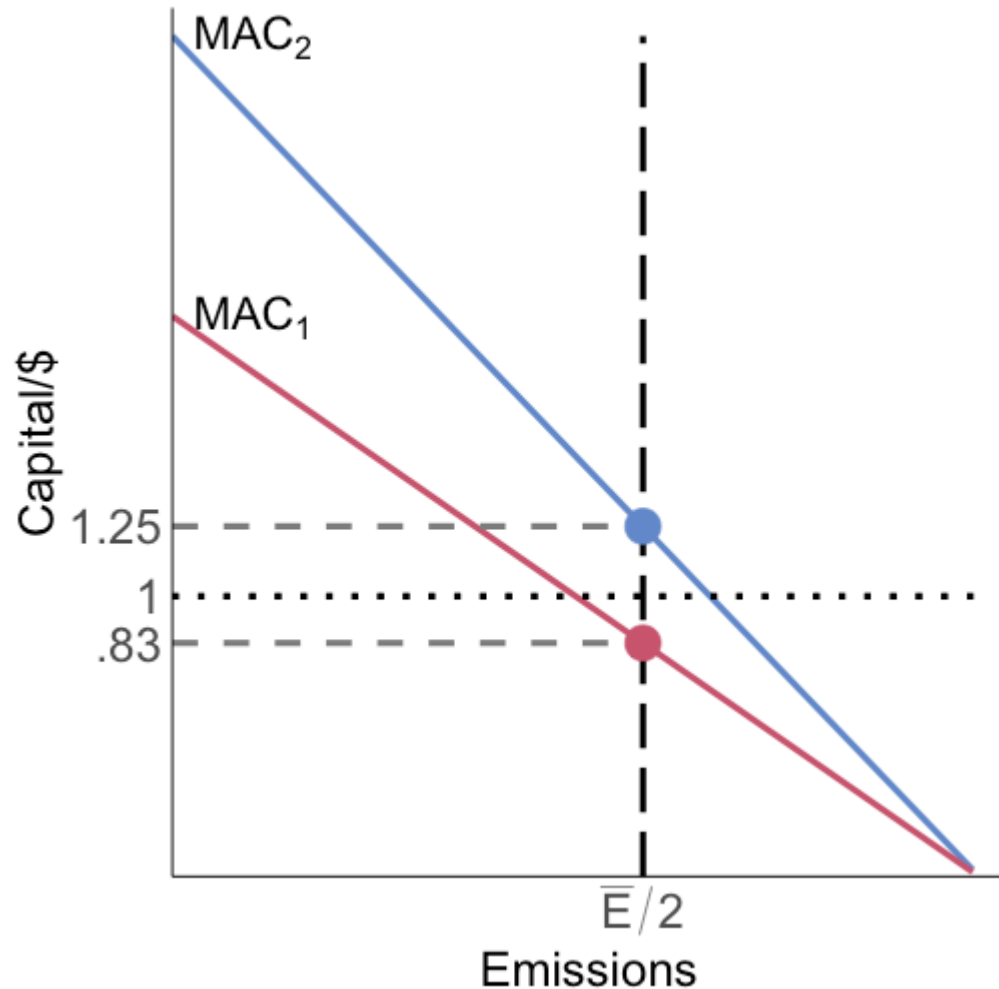


We can reduce costs by increasing emissions at high MAC firm 2 and decreasing emissions at low MAC firm 1 until they are equal

This allows us to recover DWL equal to the red area

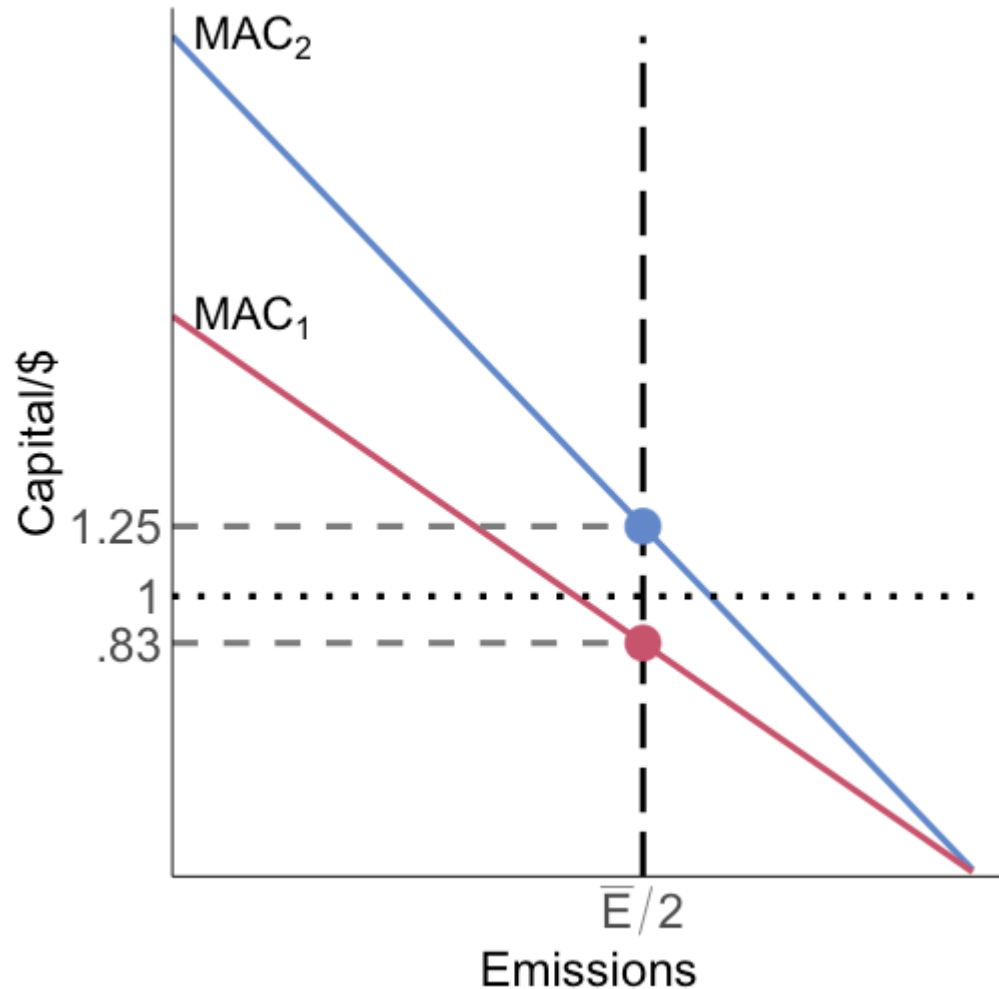
The red area is the difference in areas under MAC_2 and MAC_1 over the range of emissions changes

Tradable permits: graphical



We want to equalize MACs for cost-effectiveness, but does the permit market cause this to happen?

Tradable permits: graphical

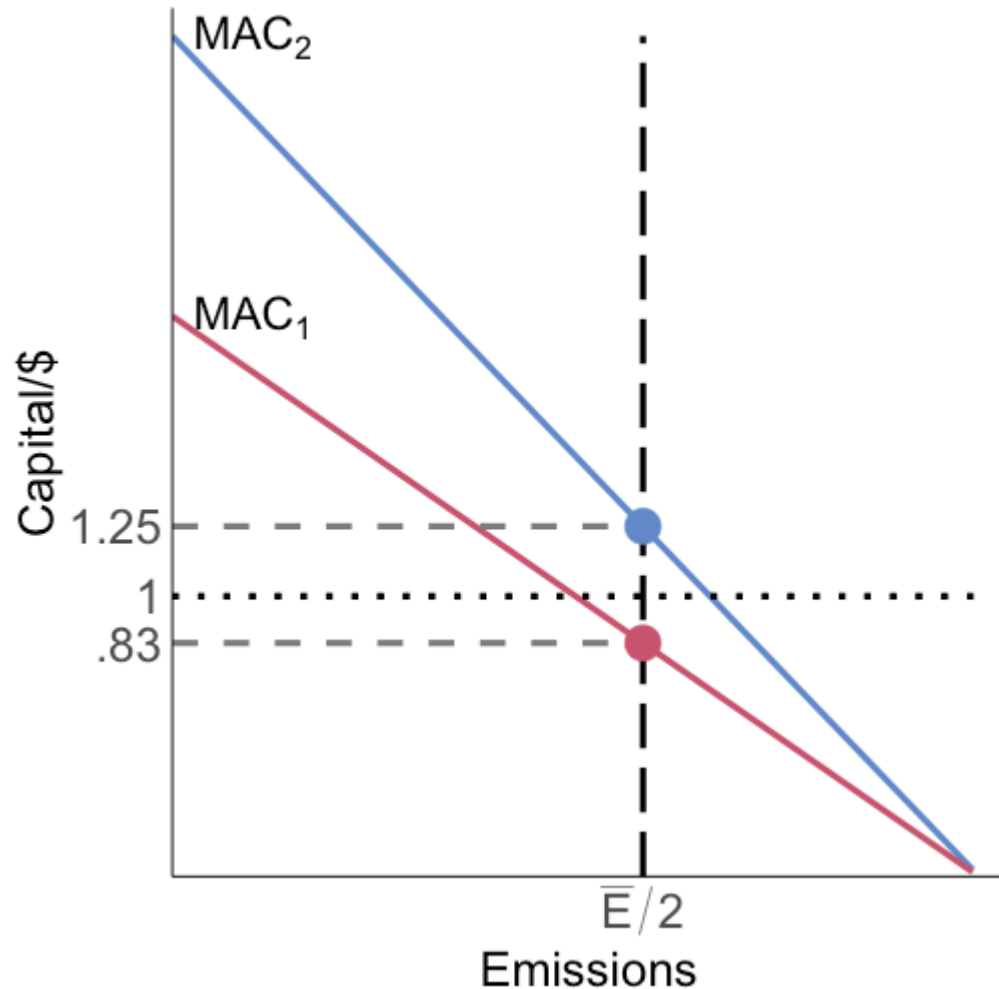


We want to equalize MACs for cost-effectiveness, but does the permit market cause this to happen?

Firm 2 is willing to pay a price up to the blue point (1.25) to be able to emit 1 more unit

Firm 1 can abate 1 more unit at cost equal to the red point (0.83)

Tradable permits: graphical

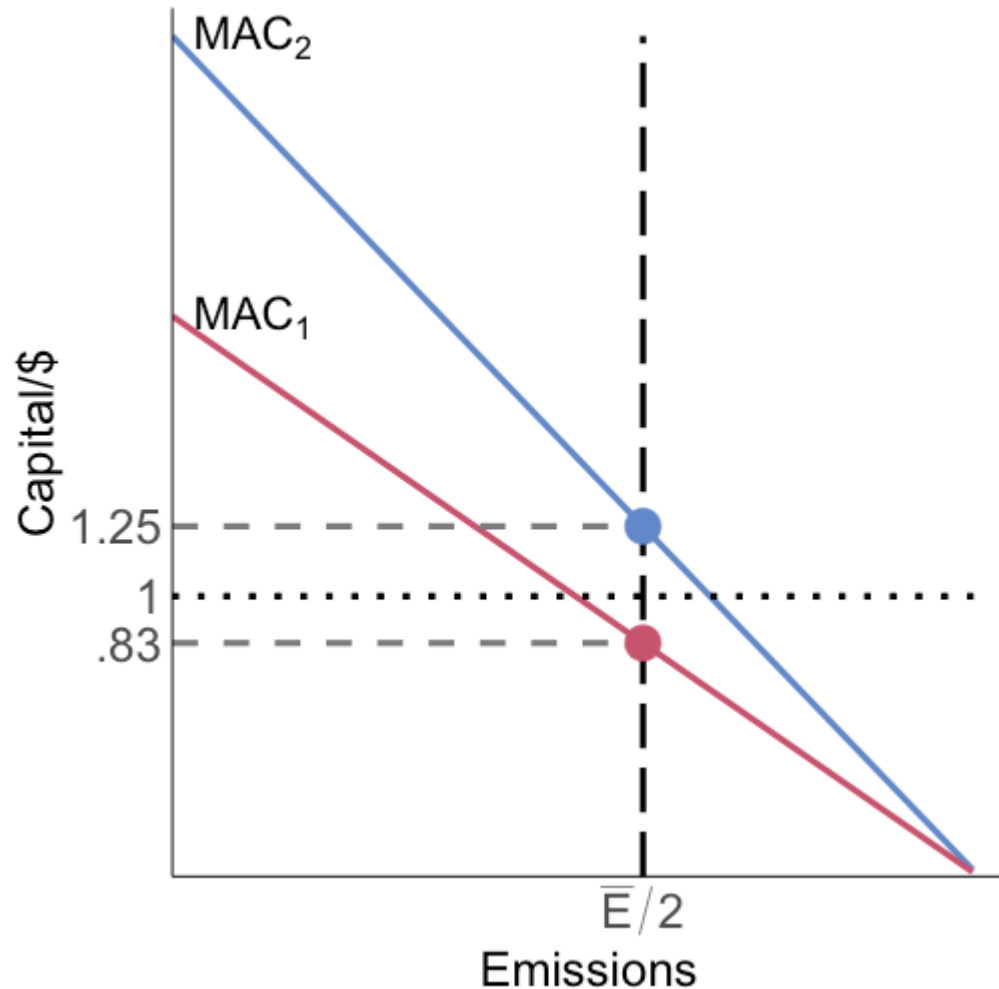


Firm 2 can buy the right to emit 1 unit of pollution from firm 1 for anywhere between 1.25 and 0.83 and **both will be better off** [very Coasean!]

These trades can be done until the MACs are equal at a value of 1

This would be the prevailing permit price in a tradable permit system

Tradable permits: graphical



An alternative way to think about it:

the prevailing permit price is the
MC of freeing up one more permit,
the MAC of the selling firm

or it is the MB of freeing up one
more permit (avoided MAC), the
MAC of the buying firm

Tradable permits: firm

We can also see this result mathematically

Suppose there is a permit price p in the competitive tradable permit market

Firms are price-takers in the permit market

Let's set up the firm problem: they want to minimize the cost of satisfying the policy

Tradable permits: firm

The firm's problem is then:

$$\min_E C(E) + pE$$

The firm's first-order condition to minimize costs is:

$$-C'(E^*) = p$$

The firm minimizes costs by choosing emissions E^* so that its MAC equals the permit price

Tradable permits: cost-effectiveness

This makes sense!

The permit price is the MC of emitting, the MAC is the MB of emitting (reduced abatement cost)

Costs are minimized when these two things are equal

Tradable permits: cost-effectiveness

What else does firm behavior tell us about permits?

If firms all set their MACs equal to p then all their MACs are equal to one another, **we have cost-effectiveness:**

$$-C'_1(E_1^*) = -C'_2(E_2^*) = \dots = -C'_N(E_N^*) = p$$

Taxes and permits both achieve cost-effectiveness

Why?

Because firms treat permit prices and a tax identically in decisionmaking

Tradable permits: cost-effectiveness

Tradable permit systems are **always** cost-effective: whatever emissions limit you set, it will be achieved at least-cost

This does not mean that it is necessarily efficient!

Tradable permits: efficiency

For efficiency, we also need $MAC = MD$

To keep things simple suppose MD is constant and the same across all firms: $MD = d$

If we can set $\bar{E}$ such that the equilibrium permit price $p = d$, then we also have efficiency

Tradable permits in practice

Knowing MD is often difficult in practice

Tradable permit systems are nice because we can just let politicians choose a $\bar{E}$ that is politically feasible, and then the permit market ensures that we get the associated emissions reductions at least-cost

What often happens in practice is $\bar{E}$ starts high, giving us a low p , and then $\bar{E}$ gets ratcheted down over time

Tradable permits: numerical example

Suppose two firms trade permits with total cap:

$$E_1 + E_2 = \bar{E} = 90$$

Abatement cost functions:

$$C_1(E_1) = \frac{1}{2}(90 - E_1)^2, \quad C_2(E_2) = \frac{1}{4}(150 - 2E_2)^2$$

So marginal abatement costs are:

$$MAC_1(E_1) = -C'_1(E_1) = 90 - E_1$$

$$MAC_2(E_2) = -C'_2(E_2) = 150 - 2E_2$$

Tradable permits: numerical example

In a competitive permit market:

$$MAC_1(E_1^*) = p, \quad MAC_2(E_2^*) = p, \quad E_1^* + E_2^* = \bar{E} = 90$$

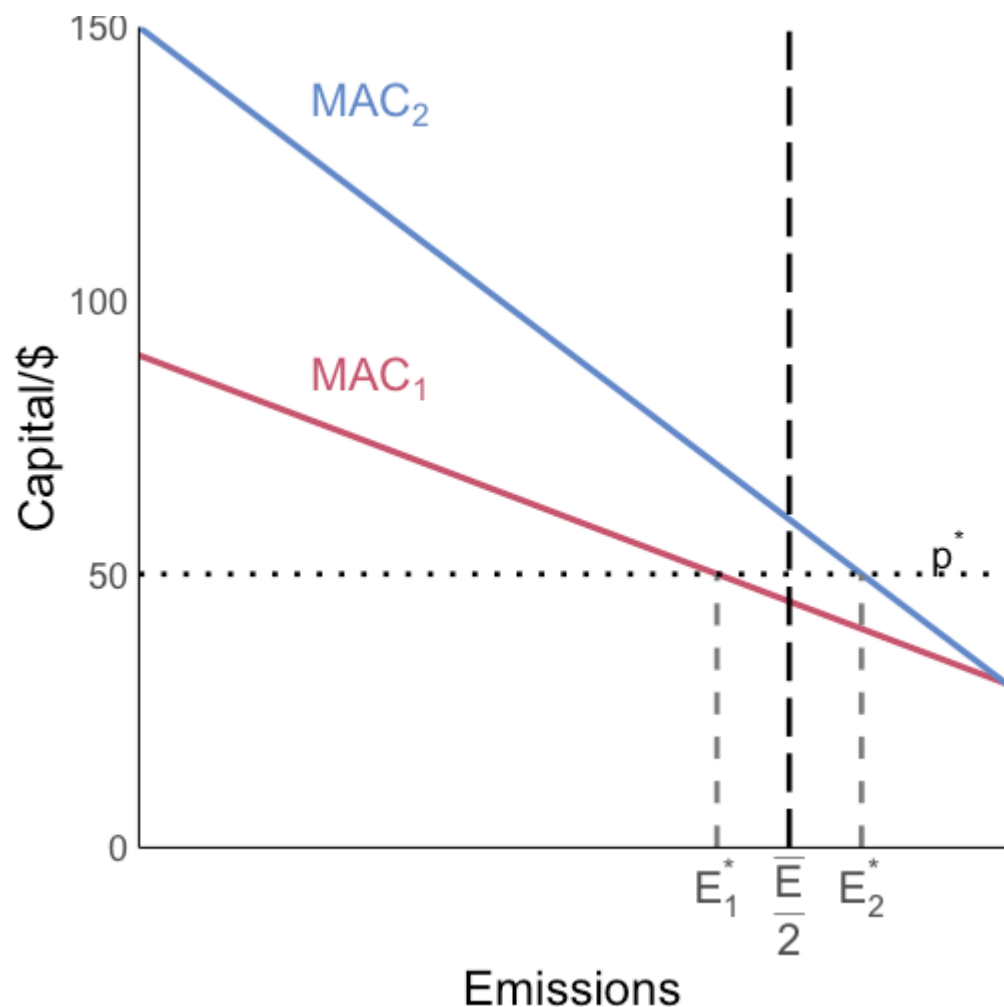
Plug in the MAC expressions:

$$90 - E_1^* = p, \quad 150 - 2E_2^* = p, \quad E_1^* + E_2^* = 90$$

$$E_1^* = 40, \quad E_2^* = 50, \quad p^* = 50$$

Firm 2 has higher MAC at an equal split, so it buys permits and emits more in equilibrium.

Tradable permits: numerical example



Numerical solution:

$$E_1^* = 40, \quad E_2^* = 50, \quad p^* = 50$$

Starting from equal allocation ($\frac{\bar{E}}{2} = 45$ each):

- firm 1 sells permits and emits less
- firm 2 buys permits and emits more

Political economy of permits

Firms are often more willing to accept a higher p than τ because they are often endowed with (some) permits for free

This means that *in practice*, we might expect to get greater emissions reductions under a permit system than a tax because of these political economy reasons

This is one of the key reasons the 1990 CAA amendments were able to be passed

Permit market challenges

How we do initially allocate permits?

Auction charge firms for each permit they hold, let price be set by marketplace, revenues can be used in other ways by the government, auction price will be the same as a Pigouvian tax

Lottery: Randomly assign permits

Grandfathering: give permits to existing firms based on historical emissions

Permit auctions

Recall the firm's problem when a permit costs p :

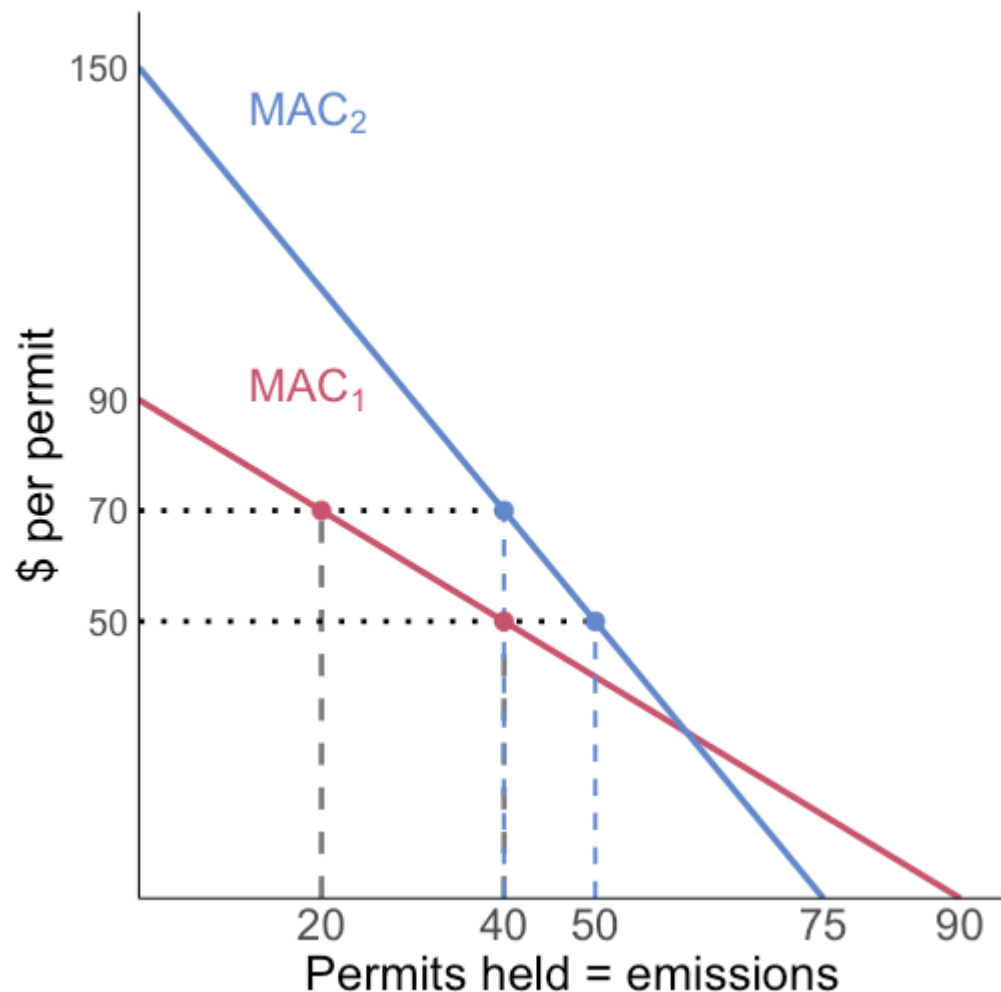
$$\min_E C(E) + pE \quad \Rightarrow \quad MAC(E) = p$$

Flip it around: at price p , the firm wants permits up to where $MAC(E) = p$

The MAC curve is the firm's demand curve for permits

Each permit lets the firm emit one more unit, so it is worth the abatement cost it avoids

Permit auctions: MAC is permit demand



Firm 1: $MAC_1 = 90 - E_1$

At $p = 70$: wants 20 permits

At $p = 50$: wants 40 permits

Firm 2: $MAC_2 = 150 - 2E_2$

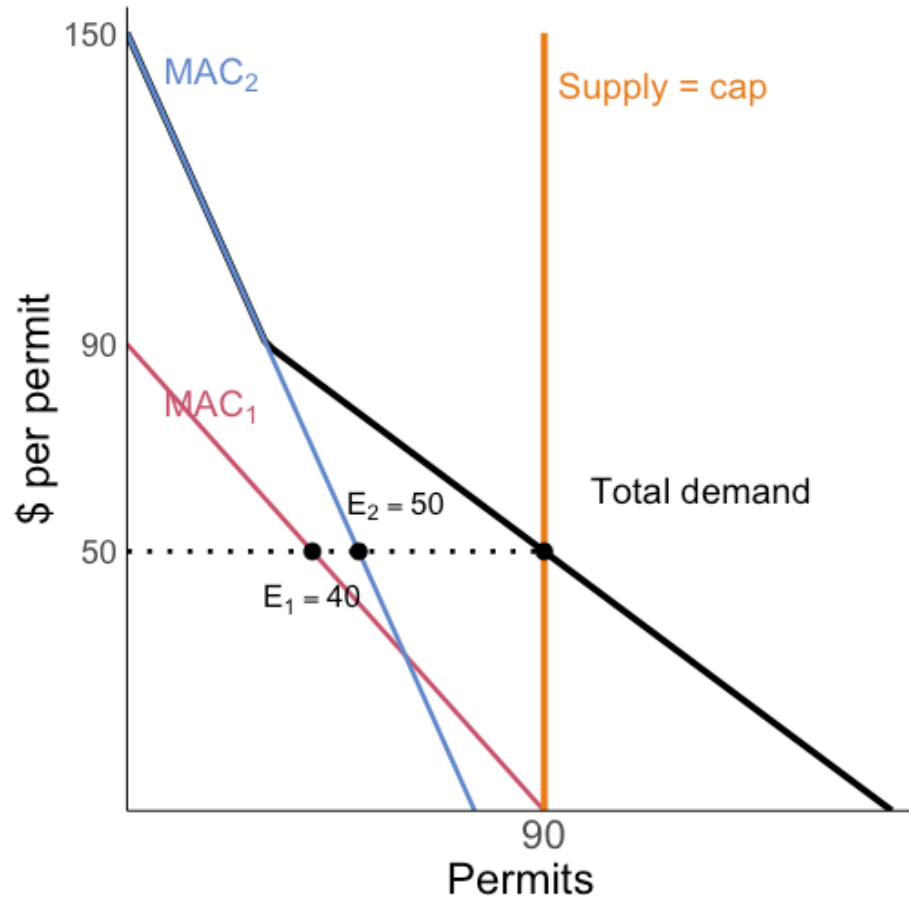
Setting $MAC_2 = p$ gives

$$E_2 = 75 - p/2$$

At $p = 70$: wants 40 permits

At $p = 50$: wants 50 permits

Permit auctions: supply meets demand



Supply: the cap, $\bar{E} = 90$

Firm 2: $MAC_2 = 150 - 2E_2$

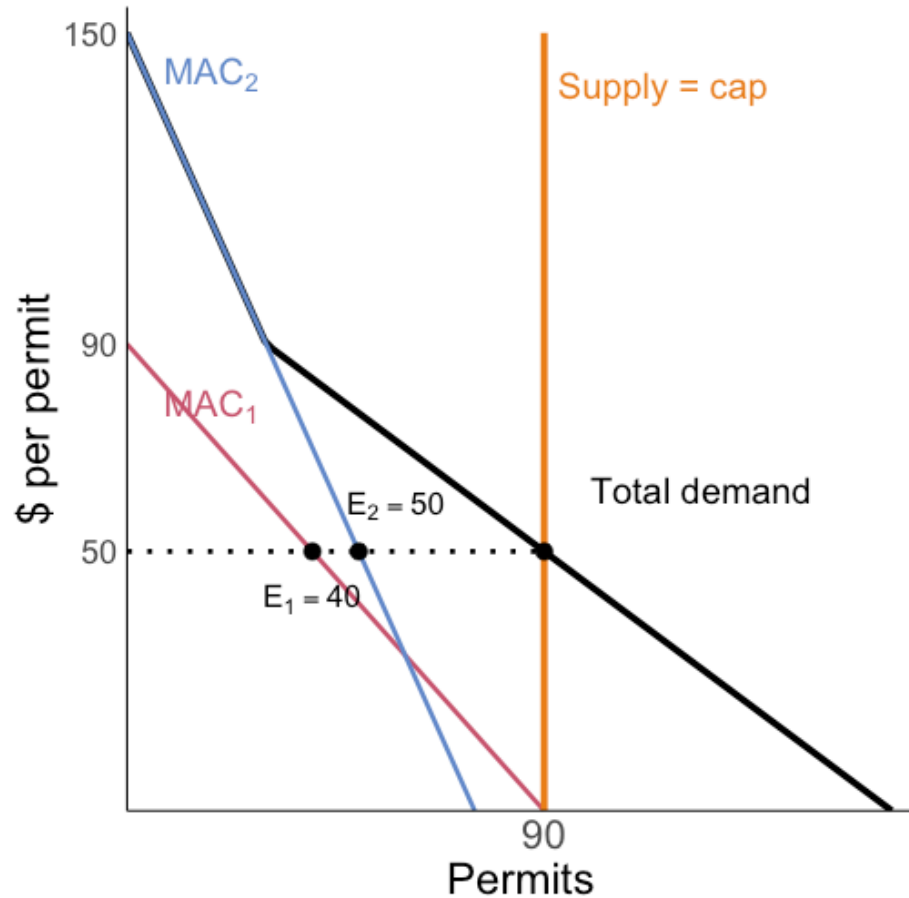
At price p , it wants $75 - p/2$ permits

Demand: add up the firms' demands¹

$$(90 - p) + (75 - p/2)$$

¹ For $p \leq 90$; above that, firm 1 wants no permits.

Permit auctions: supply meets demand



Supply: the cap, $\bar{E} = 90$

Demand for $p \leq 90$:

$$(90 - p) + (75 - p/2)$$

Set equal: $165 - 1.5p = 90$, so
 $p = 50$

$E_1 = 40$, $E_2 = 50$: **the trading equilibrium**

Permit auctions: the rules

Major permit auctions (RGGI, California-Quebec, EU ETS) are **uniform-price sealed-bid** auctions. These are our Round 1 rules:¹

1. Each firm privately bids the most it would pay for each permit
2. The regulator ranks all bids from highest to lowest
3. The top $\bar{E}$ bids win
4. **Every** winner pays the same price: the lowest winning bid

¹ The EU ETS and California-Quebec also charge the lowest winning bid; RGGI charges the highest losing bid; EPA's Acid Rain Program auctions charged each winner its own bid.

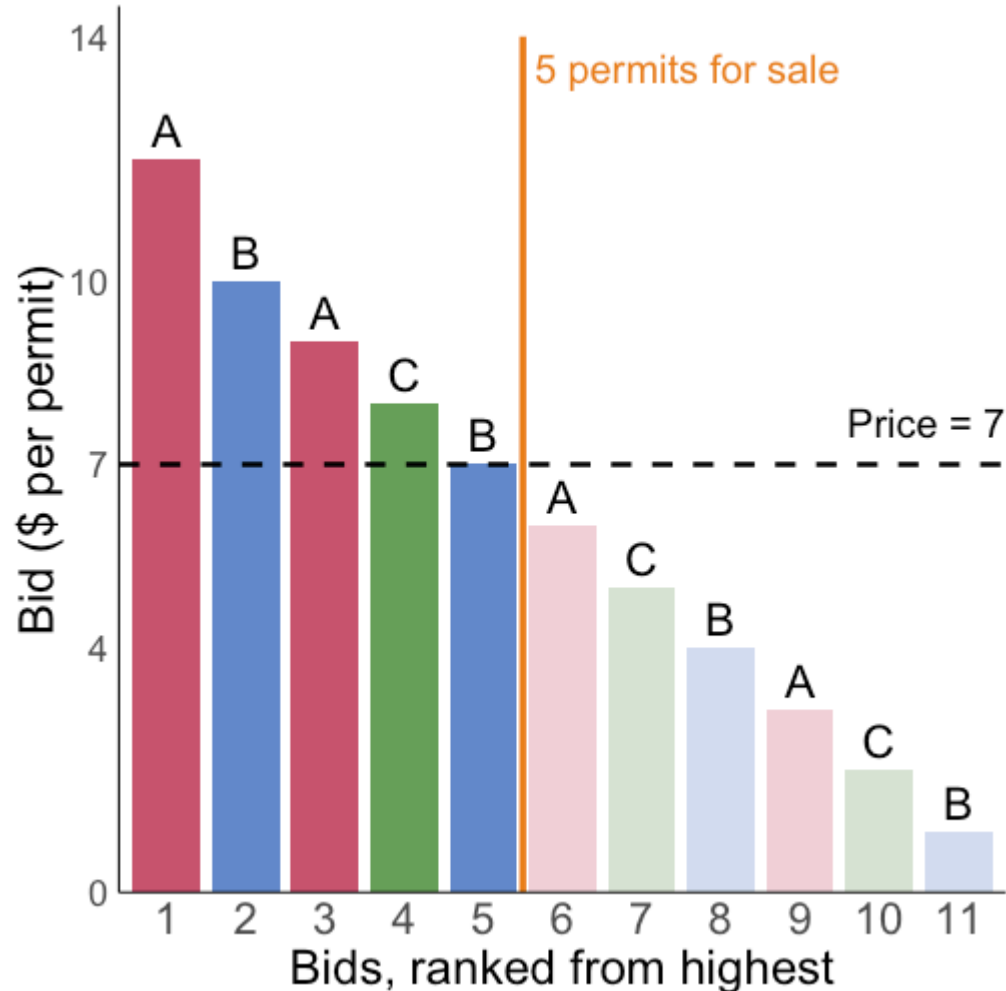
Permit auctions: example

Three firms bid for **5 permits**:

	Permit 1	Permit 2	Permit 3	Permit 4
Firm A	12	9	6	3
Firm B	10	7	4	1
Firm C	8	5	2	

Which bids win? What is the price? What does each firm pay?

Permit auctions: example



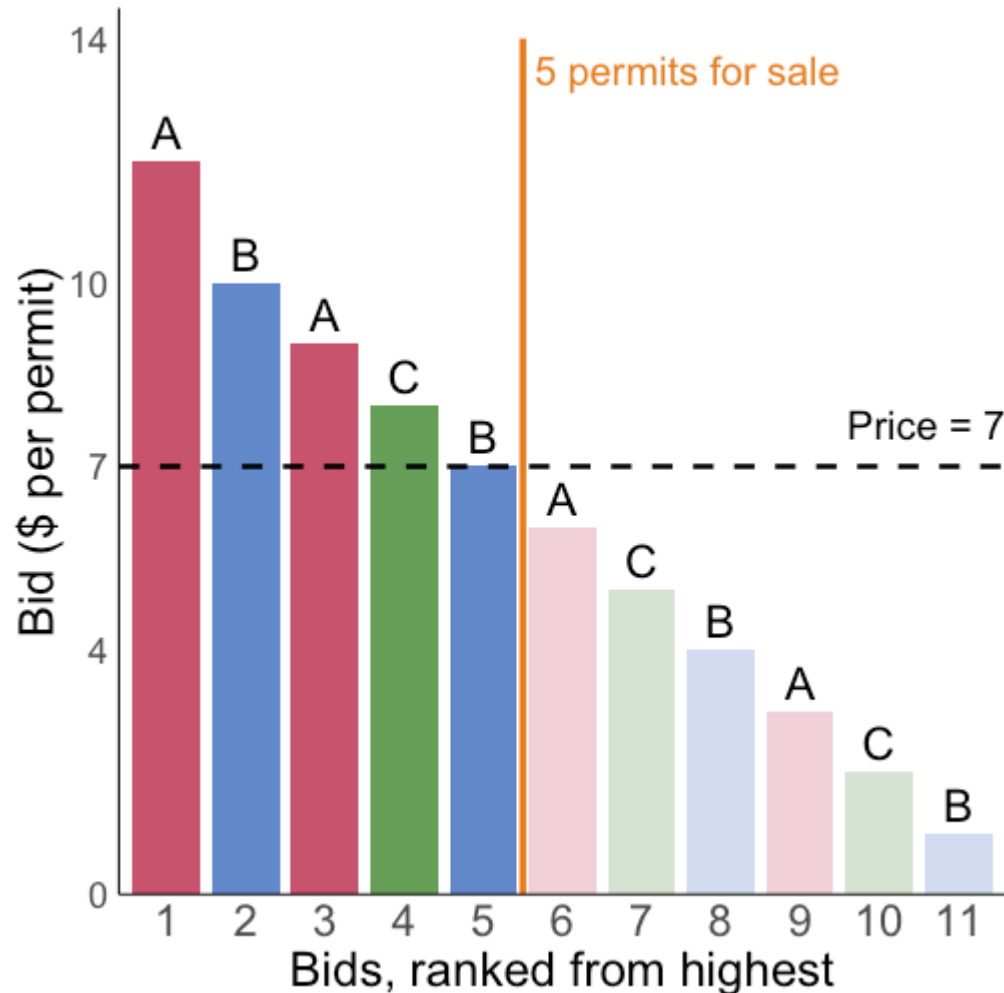
Top 5 bids win: 12, 10, 9, 8, 7

Price = lowest winning bid = 7

All winners pay only 7 per permit, even when they bid higher

Your bid mostly decides whether you win a permit; the price is the lowest winning bid, usually someone else's

Permit auctions: the gain to the firm



A whole permit avoids one unit of abatement. Its value is the **area under MAC** over that unit:

$$\text{Gain} = \text{avoided abatement cost} - p$$

If a permit saves 12 in abatement cost and costs 7, the firm gains **5**

The bars show submitted bids

Let's trade: the permit market game

The permit market game: auctions and trading

Round 1: uniform-price auction. Every winning permit costs the lowest winning bid

Round 2: pay-as-bid auction. Each winning permit costs your bid, with a tighter cap

After **each auction**: your MAC shifts, then the open market begins

Round 3: transaction costs. Restart Round 2 trading: same starting permits and post-shock MACs, plus a **\$3 buyer fee per permit**

No banking or borrowing

The permit market game: cost shocks

In Rounds 1 and 2, your MAC shifts **up or down by \$2 or \$4** when trading opens. Its slope stays the same

For example, $MAC = 24 - 2E$ becomes $26 - 2E$ or $22 - 2E$

The class has balanced upward and downward shifts. With an odd number of firms, one has no shift

The caps and shifts keep cost-effective emissions in **whole permits**

Round 3 keeps Round 2's realized MACs

The permit market game: what should happen?

- Different bidding incentives under uniform-price and pay-as-bid auctions
- Trades from low-MAC to high-MAC firms after costs change
- Lower total abatement costs as firms trade toward the cost-effective allocation
- In Rounds 1 and 2, equal MACs when the class reaches that allocation
- In Round 3, the \$3 fee blocks trades whose abatement-cost savings are less than \$3

After trading closes, compare the MAC histogram with the cost-effective price line

The permit market game: scoring

$$\text{Score} = \text{avoided abatement cost} + \text{sales} - \text{purchases} - \text{fees}$$

Avoided cost is relative to abating all baseline emissions, using your **post-shock MAC**. Purchases include auction payments

Leaderboard points = your score minus your firm's benchmark score, summed over rounds

The benchmark values your cost-effective permits at the no-fee price.

Round 3 reuses Round 2's auction payments and deducts \$3 per permit bought

The permit market game: class cost reductions

$$100 \times \frac{C_{\text{start}} - C_{\text{end}}}{C_{\text{start}} - C^*}$$

- C_{start} : total abatement cost at the auction allocation
- C_{end} : total abatement cost after trading
- C^* : cost-effective total abatement cost

All three use the same **post-shock MACs** and exclude permit payments

100% means the no-fee gap closed. A zero initial gap has no percentage

Round 3 also reports **abatement cost + transaction costs**

The permit market game: bids and orders

Auction: enter one price per permit, in any order. The game ranks bids highest to lowest

Open market: submit a buy or sell order with a price and a **positive whole-number quantity**

In Rounds 1 and 2, a match trades at the **waiting order's price**

In Round 3, buy limits **include the \$3 fee**; asks specify seller receipts. A buy at 8 matches a sell at 5: buyer pays 8, seller gets 5

The instructor closes each phase to show results before starting the next

Join the permit market game

Scan to open the student login, then join with your team name



Or open aem4510.ivanrudik.com/permits

After the permit game

Debrief: monopsony power in the auction

Monopsony power: a large buyer can influence the price it pays by reducing its demand

In a uniform-price auction, lowering a bid that sets the price can lower payments on **all** the buyer's winning permits

For example, A cuts its second bid from 9 to 6. The price falls to 6, and A still wins 2 permits, paying 12 instead of 14

This **demand reduction** uses monopsony power. It can leave high-value buyers with too few permits

Debrief: why trade after an auction?

If bids equal the abatement costs each whole permit saves, the auction reaches the cost-effective allocation

In our game, MACs shift after each auction, creating new gains from trade

Firms whose MAC rose can buy from firms whose MAC fell. Trading can also correct an auction's misallocation

Debrief: pay as you bid

EPA's Acid Rain Program auctions (from 1993) were **pay-as-bid**; they sold about 2.8% of allowances, and the rest were given away

In pay-as-bid, firms **shade bids down** below their willingness to pay

A lower winning bid means paying less. Bidding too low risks losing the permit

Bids can therefore differ from avoided abatement costs, leaving gains from later trading

Permit auctions: bidding and cost-effectiveness

In a uniform-price auction, many small buyers have little **monopsony power**: each bid is unlikely to set the price

Bids close to each permit's avoided abatement cost help allocate permits to the firms that value them most

When bids equal those values, the auction produces a **cost-effective allocation**

Pay-as-bid gives even small buyers a reason to shade bids down, because each winning bid sets its own payment

Permit market challenges

How do we set up trading rules?

We can decentralize trading market to cut down on transaction costs

Do trades need to be validated by central authority to ensure permit validity?

Permit market challenges

What about transactions costs?

Transactions costs create a wedge that prevents otherwise surplus-increasing trades from being made

Examples?

Search, information, bargaining, monitoring/enforcement

Lots of these costs are fixed, prohibit small trades

Permit systems and heterogeneous MDs

So far we assumed all firms faced the same MD

This is true for things like climate change, less true for things like SO_2 or NO_x

How well do permit systems perform with heterogeneous MD?

Permit systems and heterogeneous MDs

Let's think about a setting with two firms: 1 and 2

The firms have different MACs: $MAC_1 < MAC_2$ for all E

And the firms have different marginal damages as well: $MD_1 < MD_2$ for all E

How well does a permit system work?

Permit systems and heterogeneous MDs

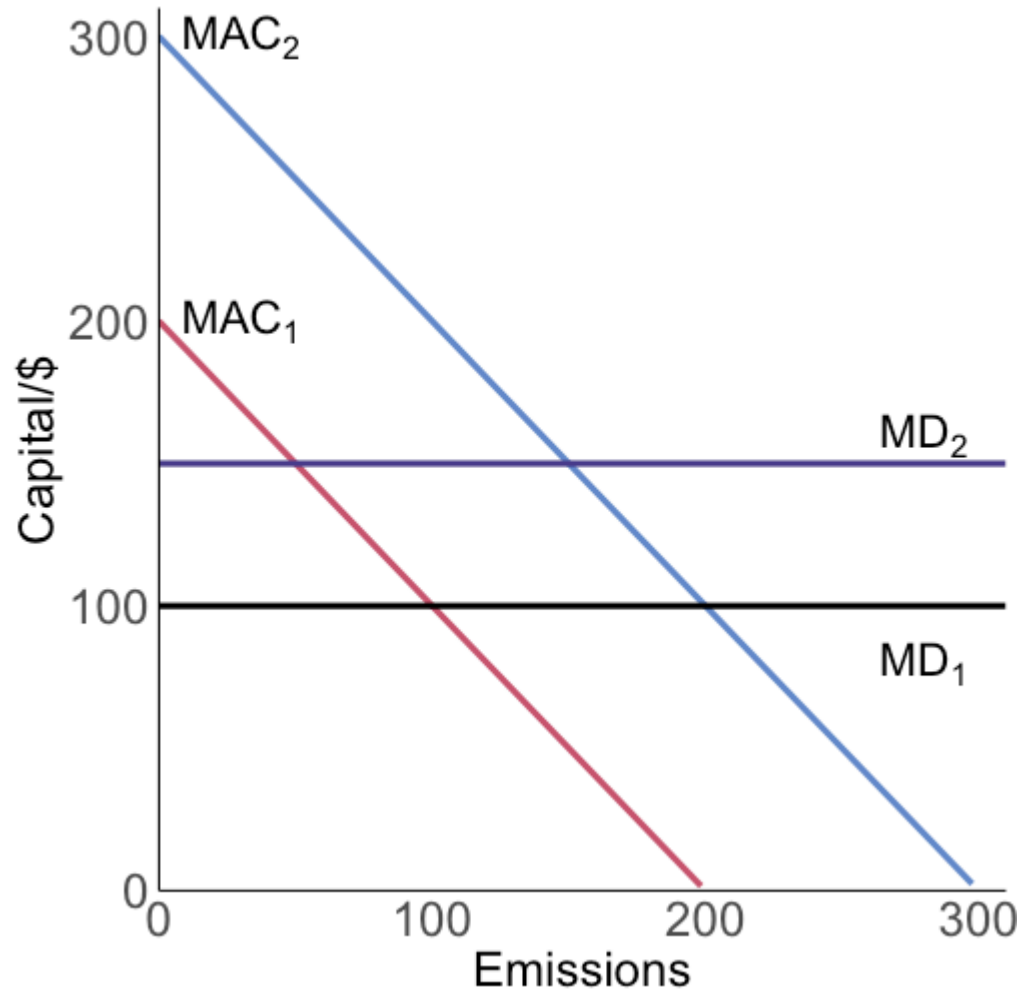
Well we know the following:

Firms set $MAC = p$ so we will have $MAC_1 = MAC_2 = p$

But for efficiency we also want $MAC = MD$: $MAC_1 = MD_1$ and $MAC_2 = MD_2$

If $MD_1 \neq MD_2$ then the permit system does **not** deliver efficiency!

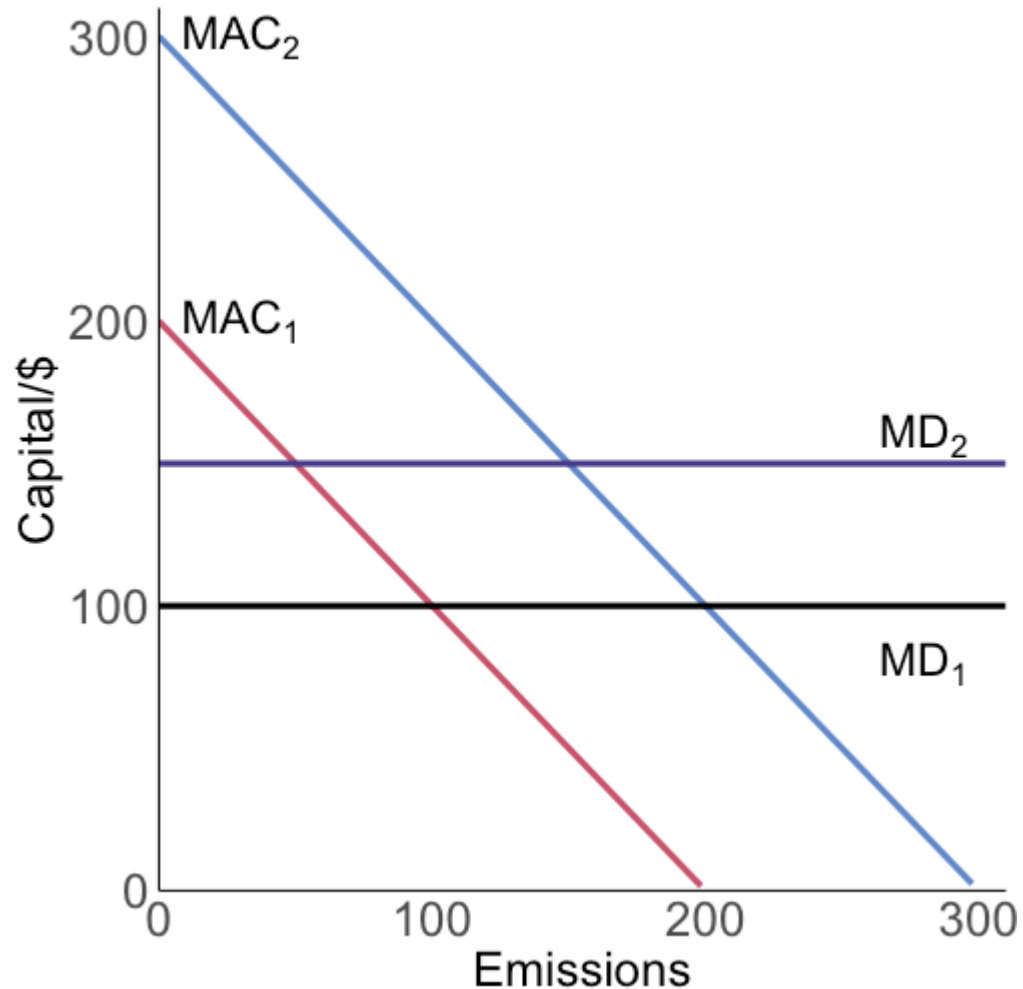
Permit systems and heterogeneous MDs: graphical



Suppose we have the two firms with different MACs and MDs:

- $MAC_1 = 200 - E_1$
- $MAC_2 = 300 - E_2$
- $MD_1 = 100$
- $MD_2 = 150$

Permit systems and heterogeneous MDs: graphical



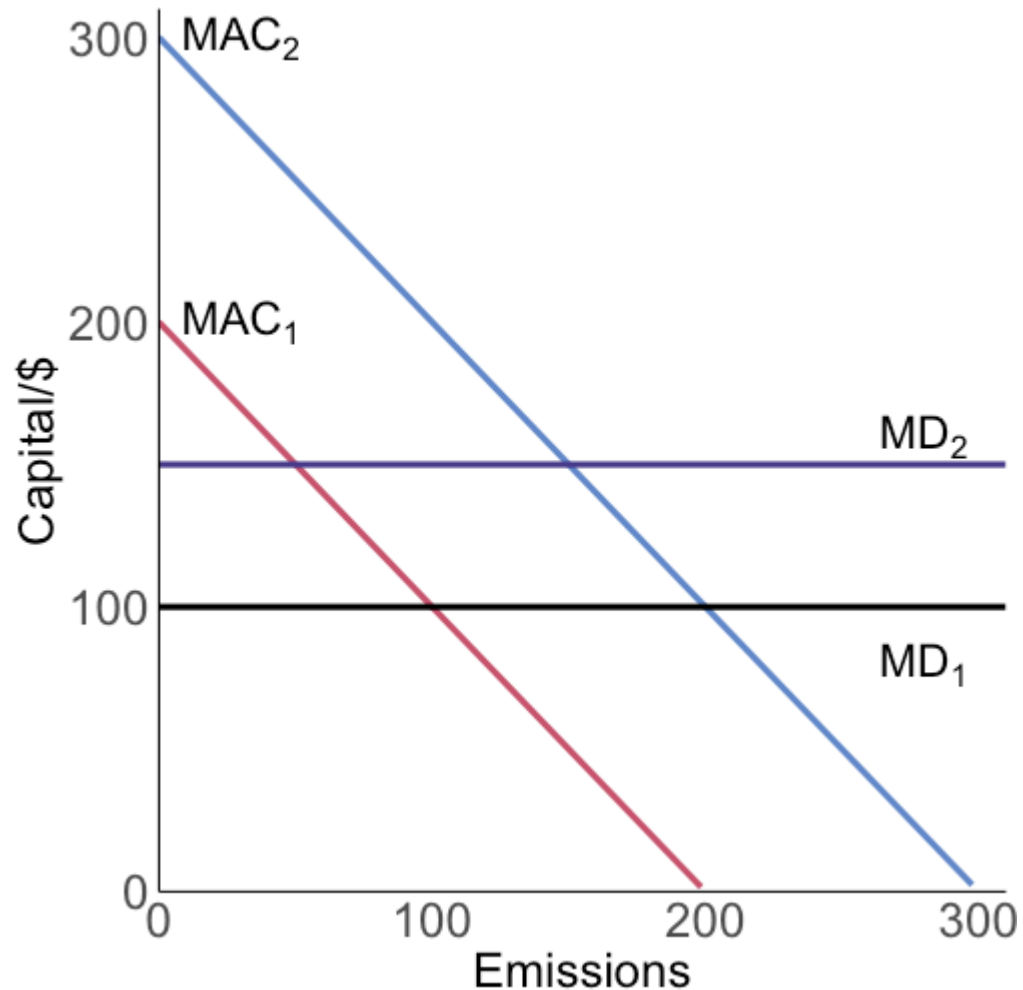
Suppose we have the two firms with different MACs and MDs:

- $MAC_1 = 200 - E_1$
- $MAC_2 = 300 - E_2$
- $MD_1 = 100$
- $MD_2 = 150$

The efficient emissions allocation is: $E^* = 250 : E_1^* = 100, E_2^* = 150$

The regulator sets $\bar{E} = 250$

Permit systems and heterogeneous MDs: graphical



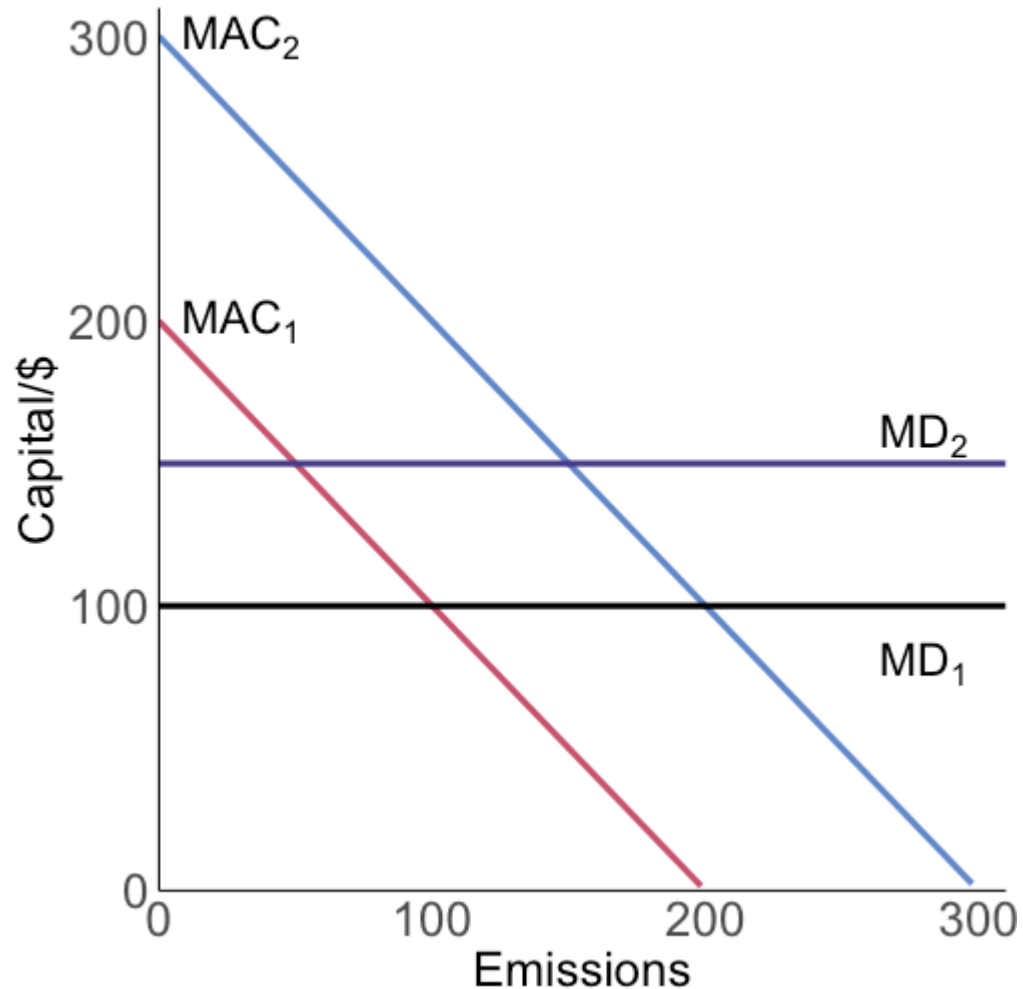
We can solve for the permit market allocation and price using:

$$MAC_1 = MAC_2 \text{ and}$$

$$E_1 + E_2 = 250$$

These two conditions tell us:

Permit systems and heterogeneous MDs: graphical



We can solve for the permit market allocation and price using:

$$MAC_1 = MAC_2 \text{ and}$$

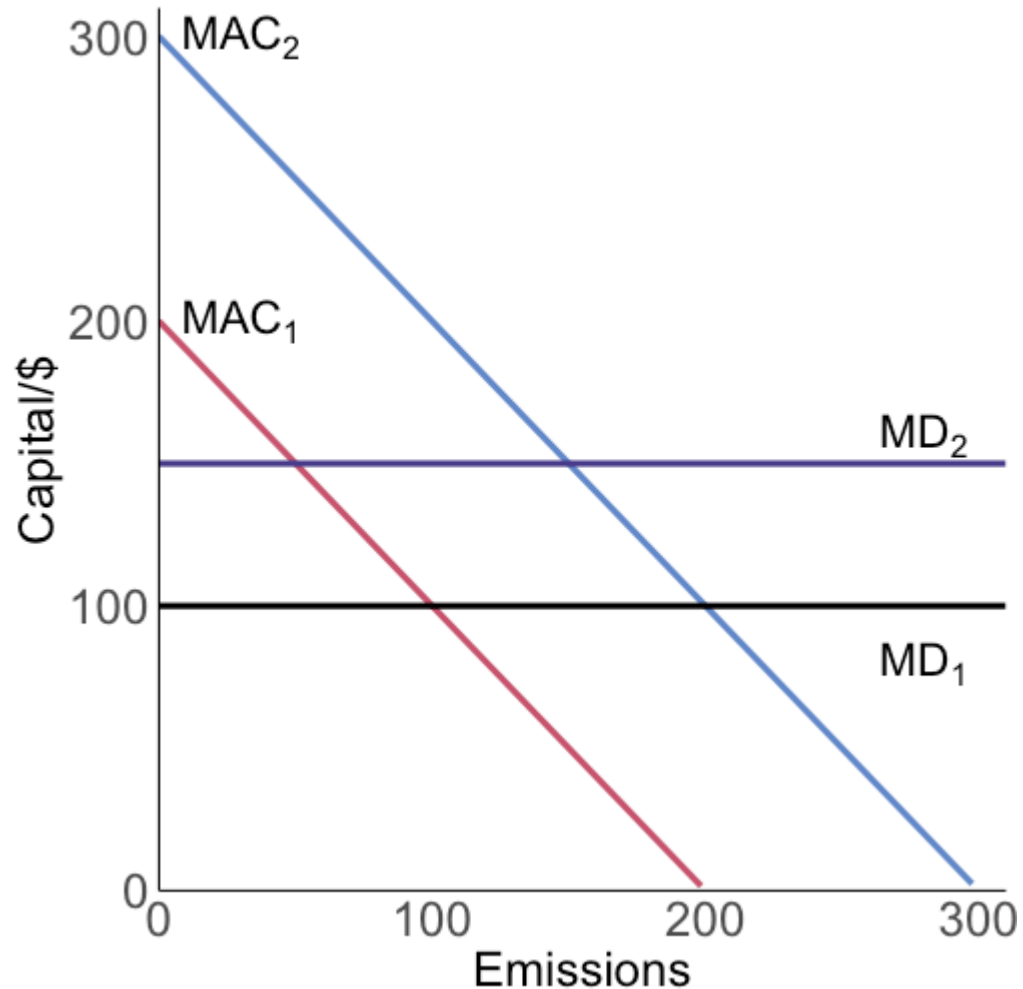
$$E_1 + E_2 = 250$$

These two conditions tell us:

$$E_1^p = 75, E_2^p = 175, p = 125$$

but efficiency is at:

Permit systems and heterogeneous MDs: graphical



We can solve for the permit market allocation and price using:

$$MAC_1 = MAC_2 \text{ and}$$

$$E_1 + E_2 = 250$$

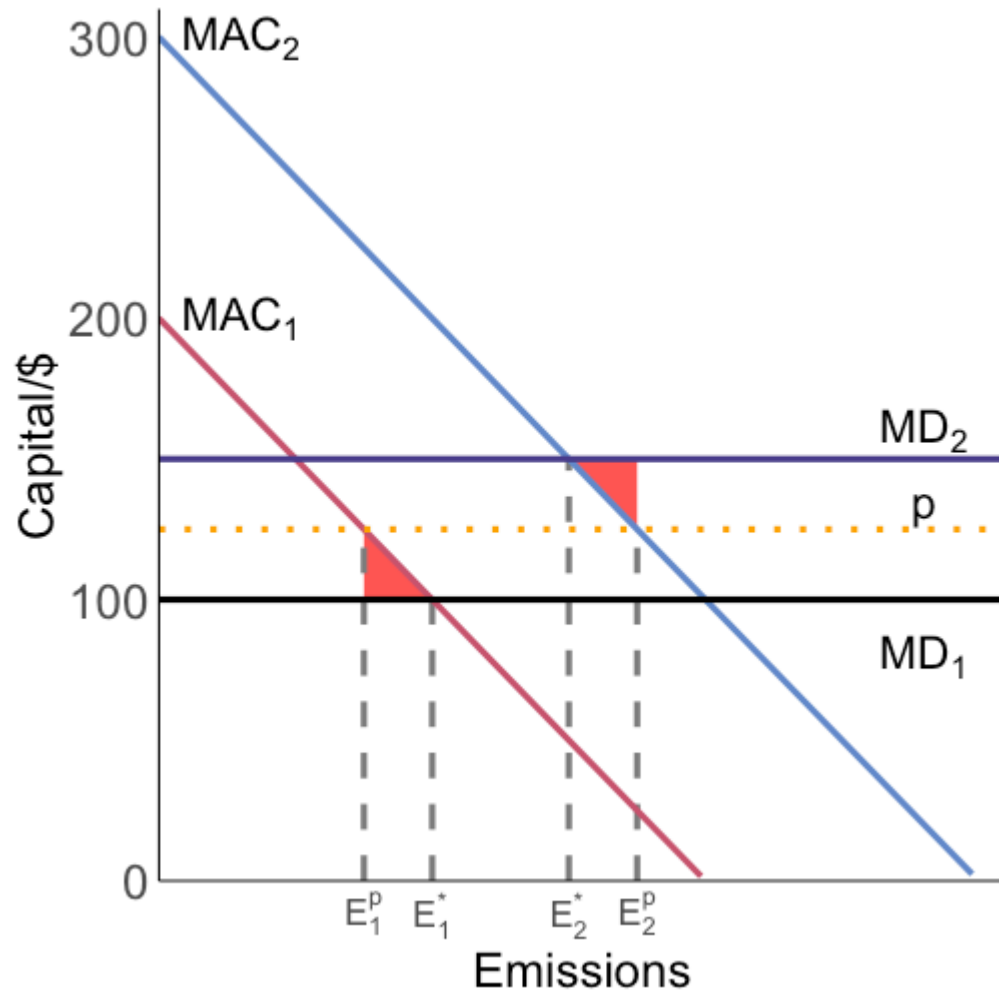
These two conditions tell us:

$$E_1^p = 75, E_2^p = 175, p = 125$$

but efficiency is at:

$$E_1^* = 100, E_2^* = 150$$

Permit systems and heterogeneous MDs: graphical



We can solve for the permit market allocation and price using:

$$MAC_1 = MAC_2 \text{ and}$$

$$E_1 + E_2 = 250$$

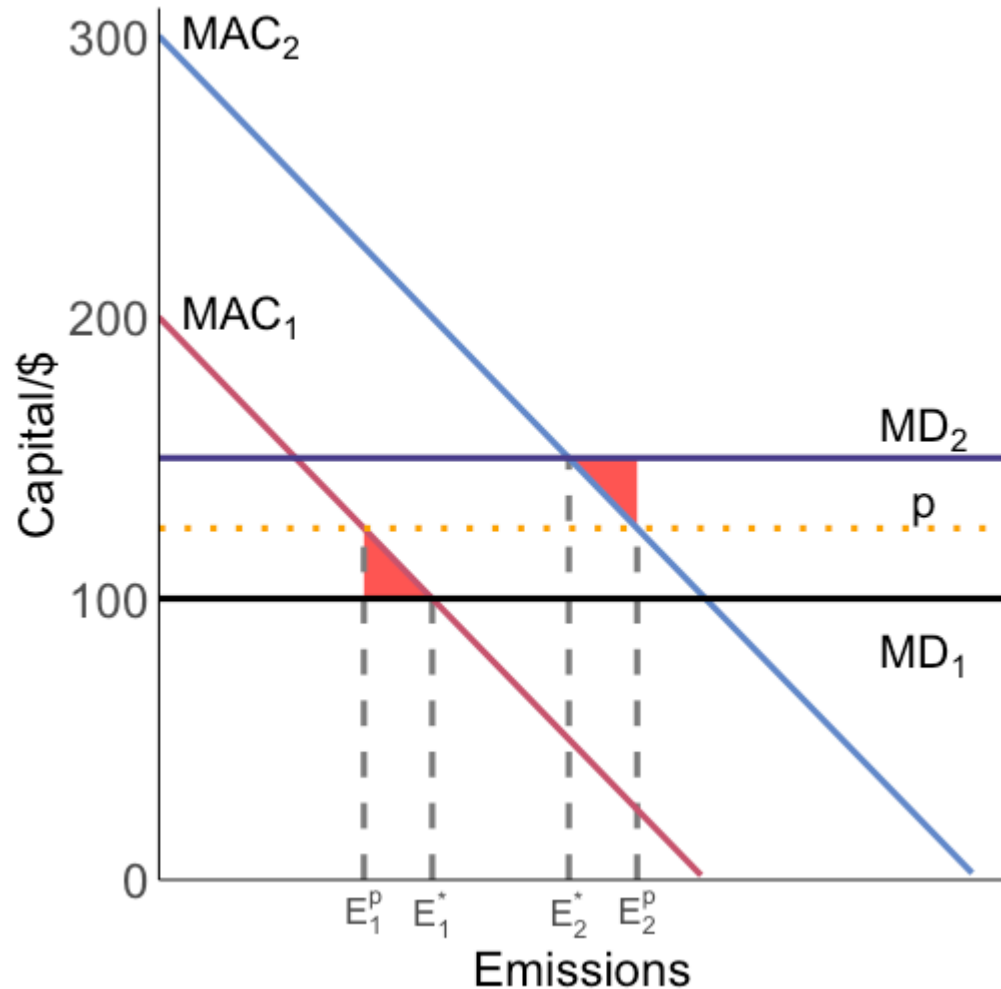
These two conditions tell us:

$$E_1^p = 75, E_2^p = 175, p = 125$$

but efficiency is at:

$$E_1^* = 100, E_2^* = 150$$

Permit systems and heterogeneous MDs: graphical

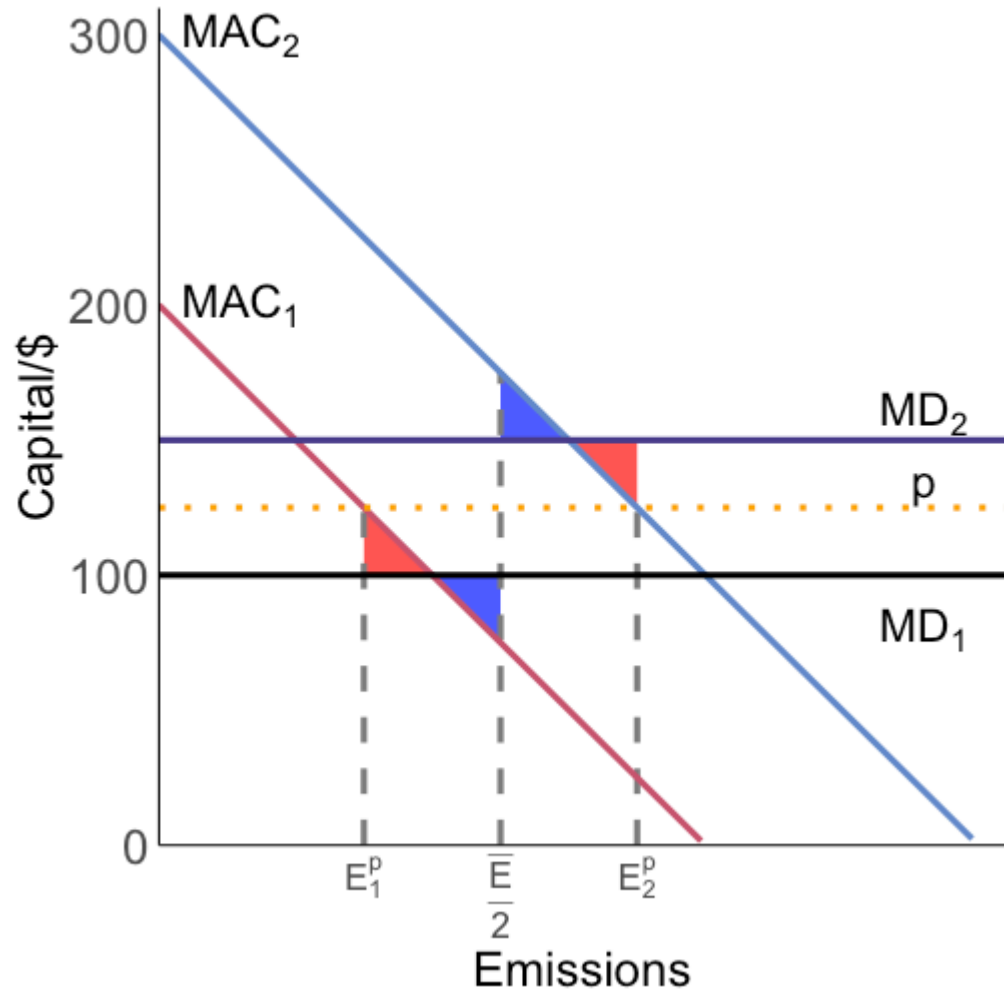


Relative to the optimal allocation, the permit system has DWL equal to the **red area**

The permit allocation is not an **efficient** allocation, but is it a **Pareto improvement** over:

1. No policy?
2. A uniform standard of $\bar{E}/2$?

Permit systems and heterogeneous MDs: graphical

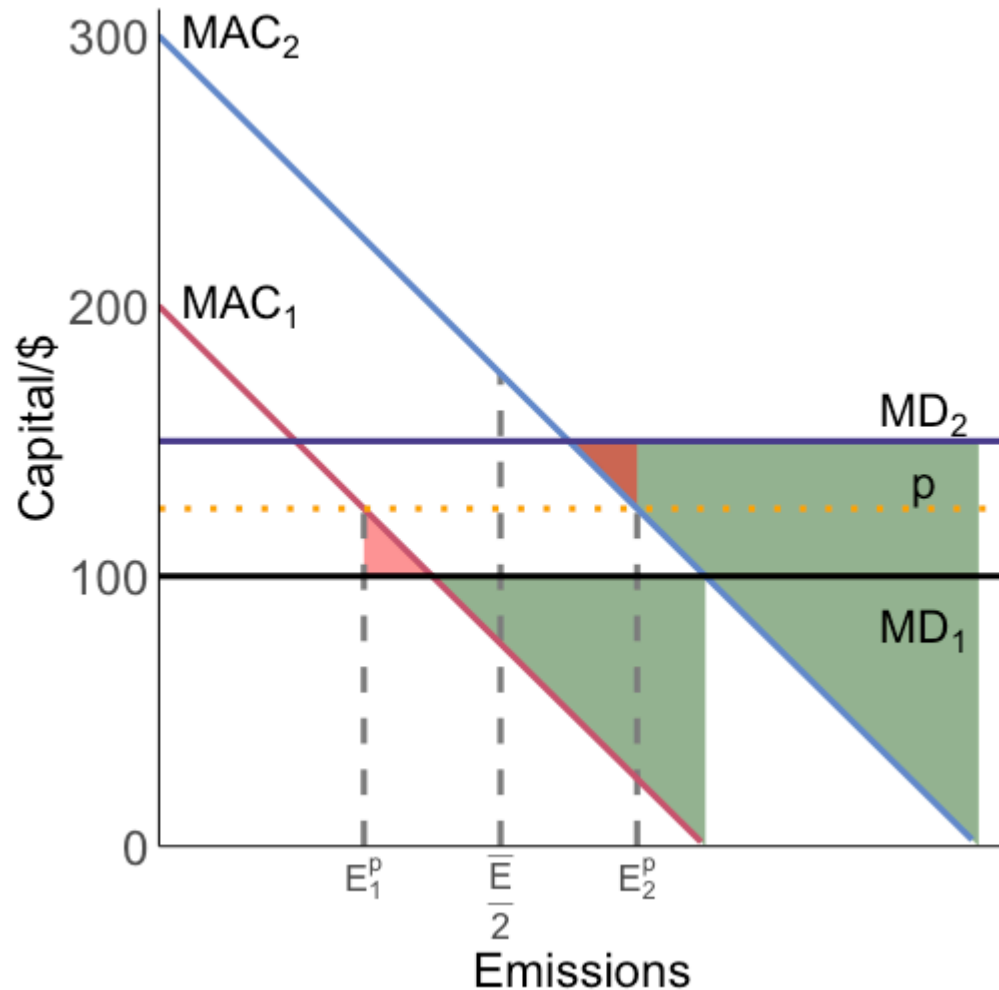


The blue area is the DWL under the uniform standard

In this specific case, a uniform standard and the permit system have the same efficiency since the red and blue areas are equal

The only difference is what kind of welfare loss is occurring where

Permit systems and heterogeneous MDs: graphical

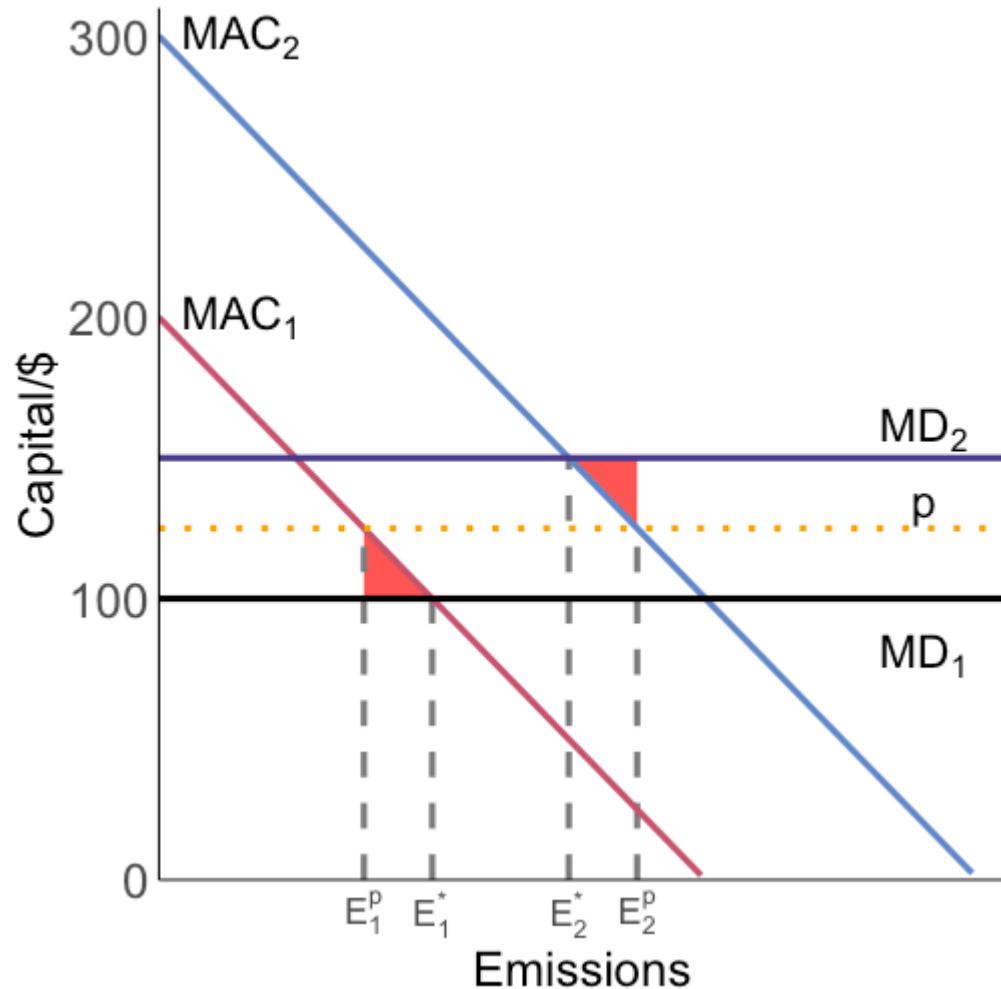


The DWL without any policy is the two large green triangles

These are clearly larger than the DWL under the permit system

The permit system can deliver a welfare improvement

Permit systems and heterogeneous MDs: graphical



What if the high MAC firm was the low MD firm?

i.e: what if the correlation between MAC and MD was **negative** instead of **positive**?

What might we expect the correlation to be?

Permit systems and heterogeneous MDs

What is the problem with permit systems and heterogeneous MD?

We want to have firms pay a price equal to their MD

Firms have different MDs but there's only one common permit price

This means we can't make all firms correctly account for their externalities

One way around this is to use **trading ratios**: firms in high damage areas need to procure more permits for the same amount of emissions

Another way is **zonal trading**: firms can only trade in similar MD areas

Trading ratios: Acid Rain Program

Below are estimates of efficient trading ratios for the Acid Rain Program

TABLE 3—TRADING RATIOS BETWEEN SOURCES AT EACH QUANTILE FOR SO₂

Quantile	Source location (county, state)	1 st	25 th	50 th	75 th	99 th	99.9 th
1 st	Josephine, OR	1.0	0.4	0.2	0.2	0.1	0.02
25 th	Polk, TX	2.5	1.0	0.6	0.4	0.1	0.1
50 th	Grant, AR	4.5	1.8	1.0	0.7	0.2	0.1
75 th	Marion, SC	6.0	2.4	1.3	1.0	0.3	0.1
99 th	Allegheny, PA	19.2	7.7	4.3	3.2	1.0	0.4
99.9 th	Hudson, NJ	49.4	19.6	11.0	8.2	2.5	1.0

Note: The trading ratio represents the number of tons from the column source required to offset one ton from the row source.

Muller and Mendelsohn (2009)

PM2.5 damages

Trading ratios are required because damages are heterogeneous across space

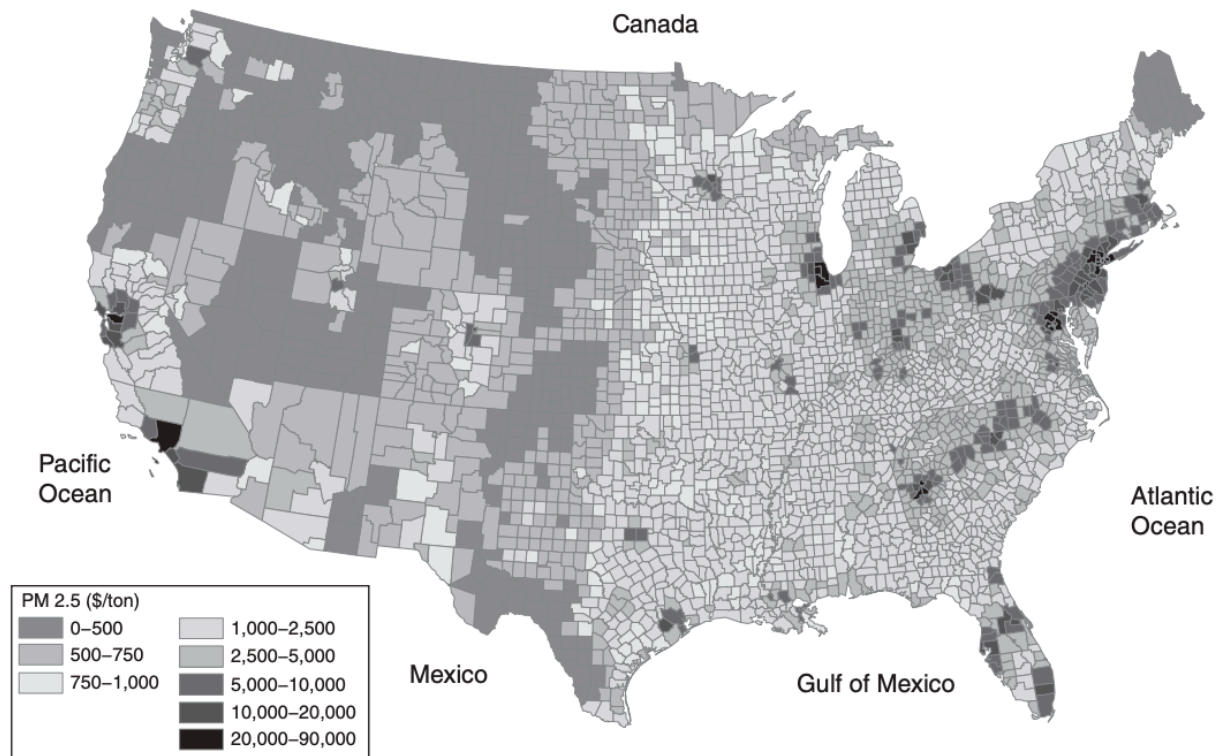
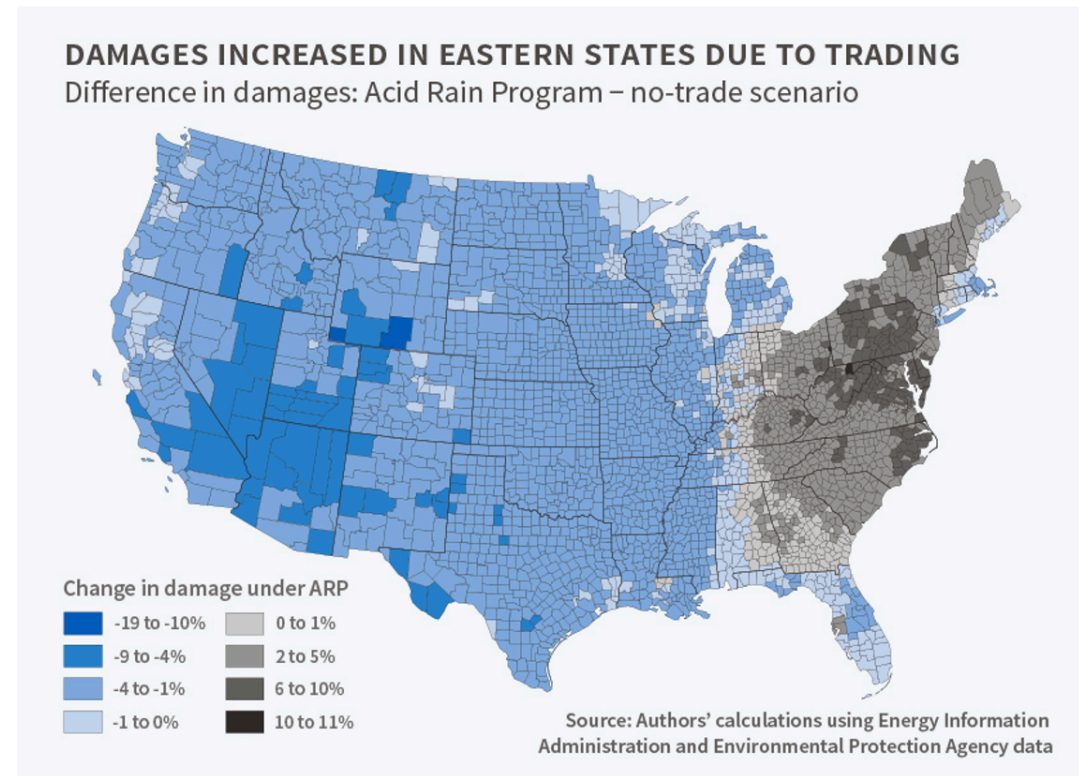


FIGURE 1. MARGINAL DAMAGE OF EMISSIONS FOR GROUND LEVEL SOURCES OF PM_{2.5} (\$/TON/YEAR)

Damages caused by ARP

The Acid Rain Program **increased** damages in the eastern US



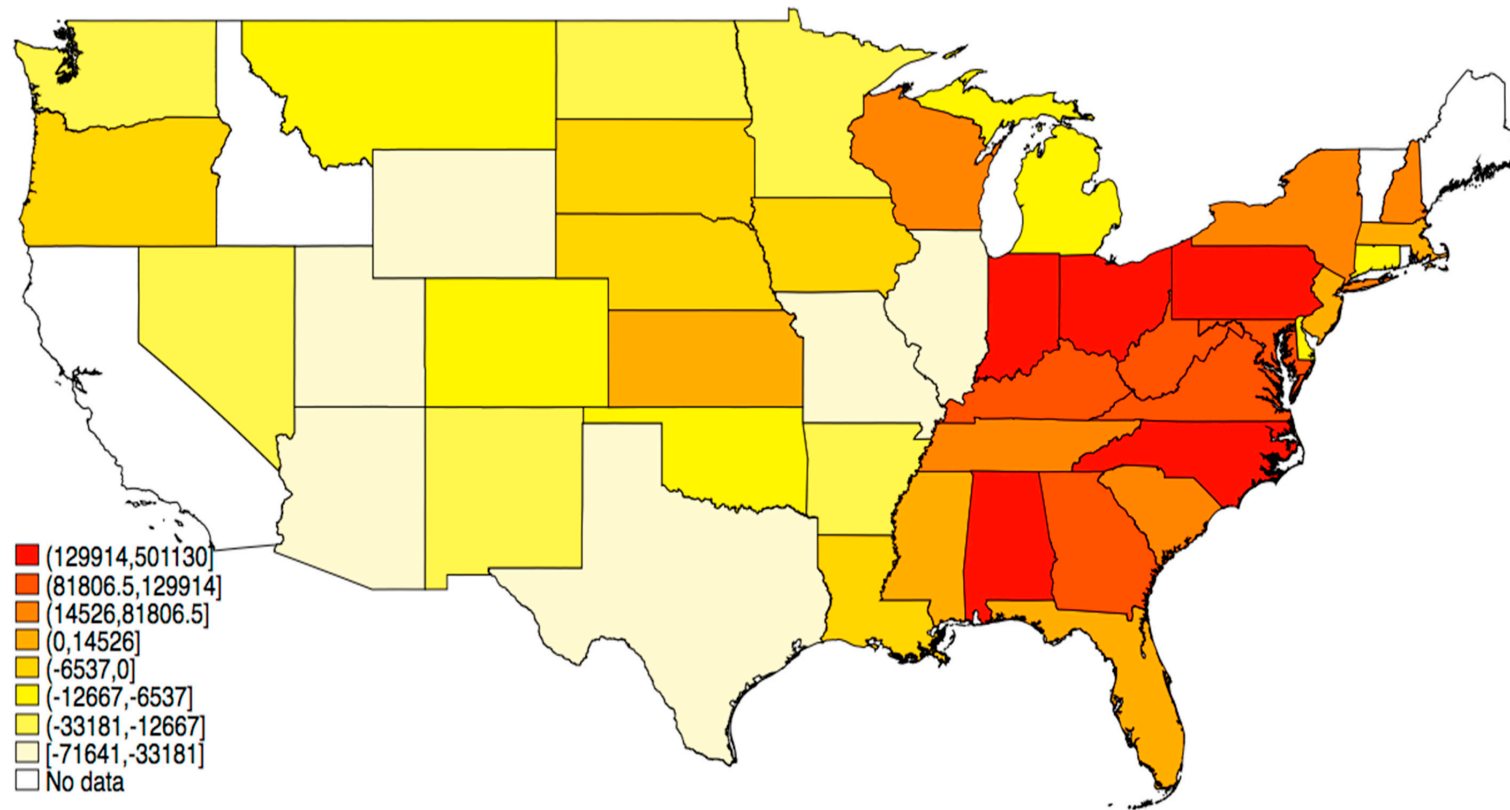
Damages caused by ARP

Chan et al. (2018) JEEM:

We also compare health damages associated with observed SO₂ emissions from all ARP units in 2002 with damages from a no-trade counterfactual. Damages under the ARP are *2.1 billion*(1995) higher than under the no-trade scenario, reflecting allowance transfers from units in the western US to units in the eastern US with larger exposed populations.

Damages caused by ARP

Redder: trading lead to greater emissions vs no trading



Zonal trading: RECLAIM

Regional Clean Air Management (RECLAIM) Program

California South Coast Air Quality Management District (SCAQMD)

Started in 1994, goal of cutting NO_x and SO_x emissions by 80% by 2003

RECLAIM is a facility-level tradable permit system

Zonal trading: RECLAIM

14% of permits allocated to power generators

Permit prices driven by electricity sector

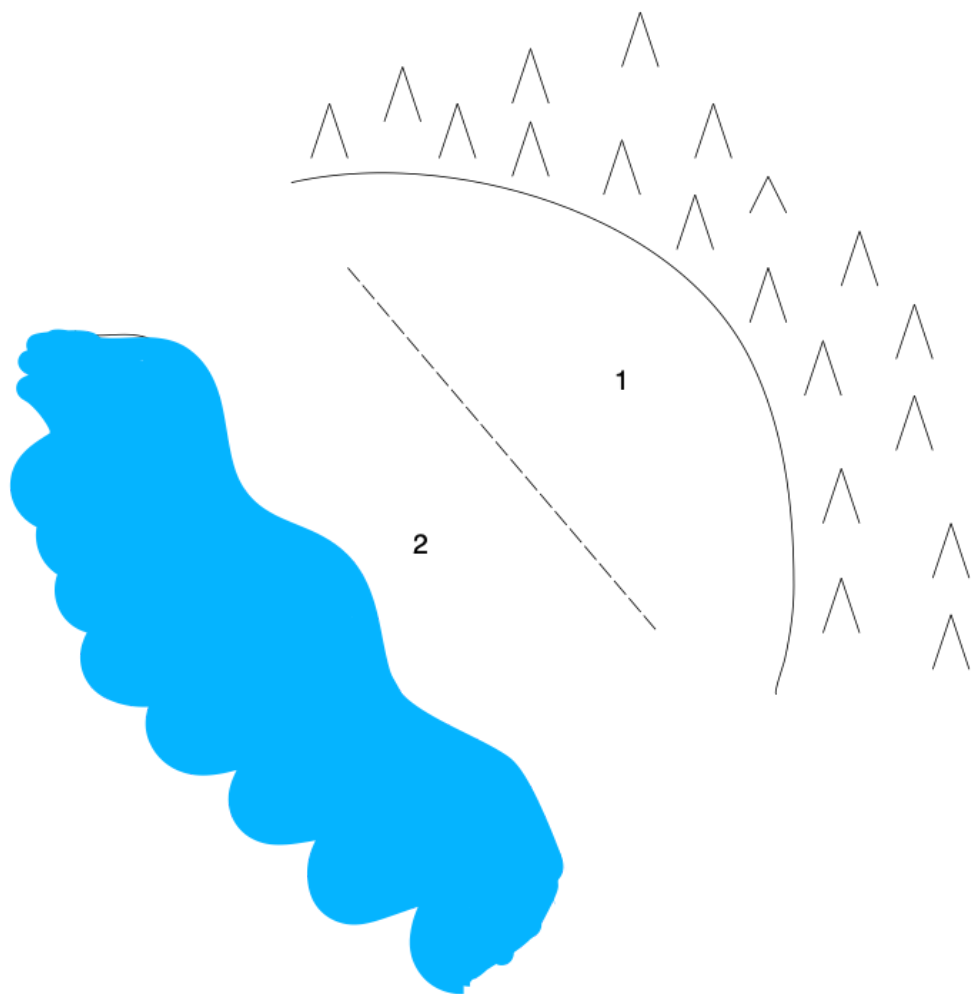
During 2000 electricity price spikes, lots of very dirty plants brought on-line to meet demand

Permit prices rose dramatically for everyone else

\$4,284 per ton of NO_x in 1999

\$39,000 per ton of NO_x in 2000

Zonal trading: RECLAIM



LA Basin has two distinct zones with very different MD's

1. Old heavy industry (high MAC) and mountains trap NO_x emissions and heat them up → smog (high MD)
2. Newer firms (low MAC) close to the ocean, breezes dissipate pollution before it can turn into smog (low MD)

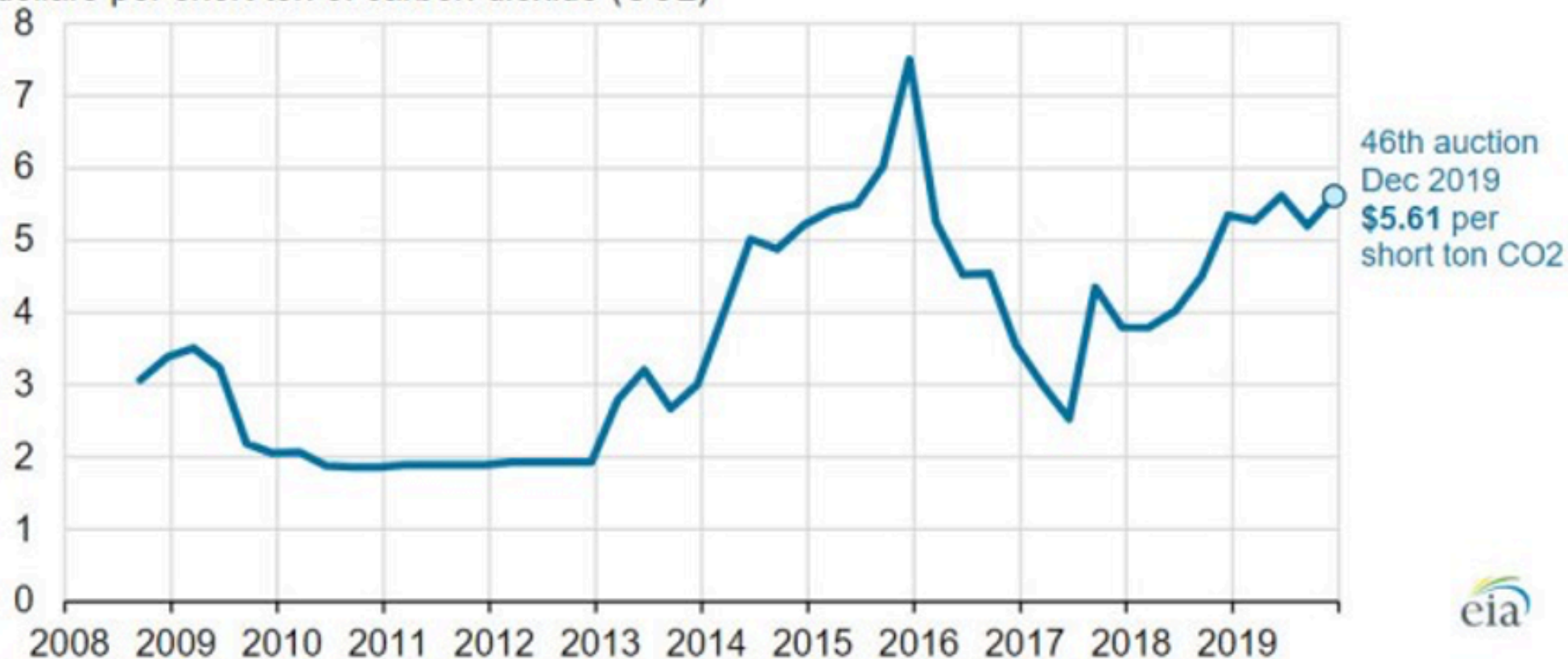
Other permit market examples

Tradable permit systems are increasingly common:

1. Acid Rain Program
2. NOx Budget Program
3. Regional Greenhouse Gas initiative
4. California AB32
5. EU Emission Trading System
6. China's National Carbon Cap and Trade

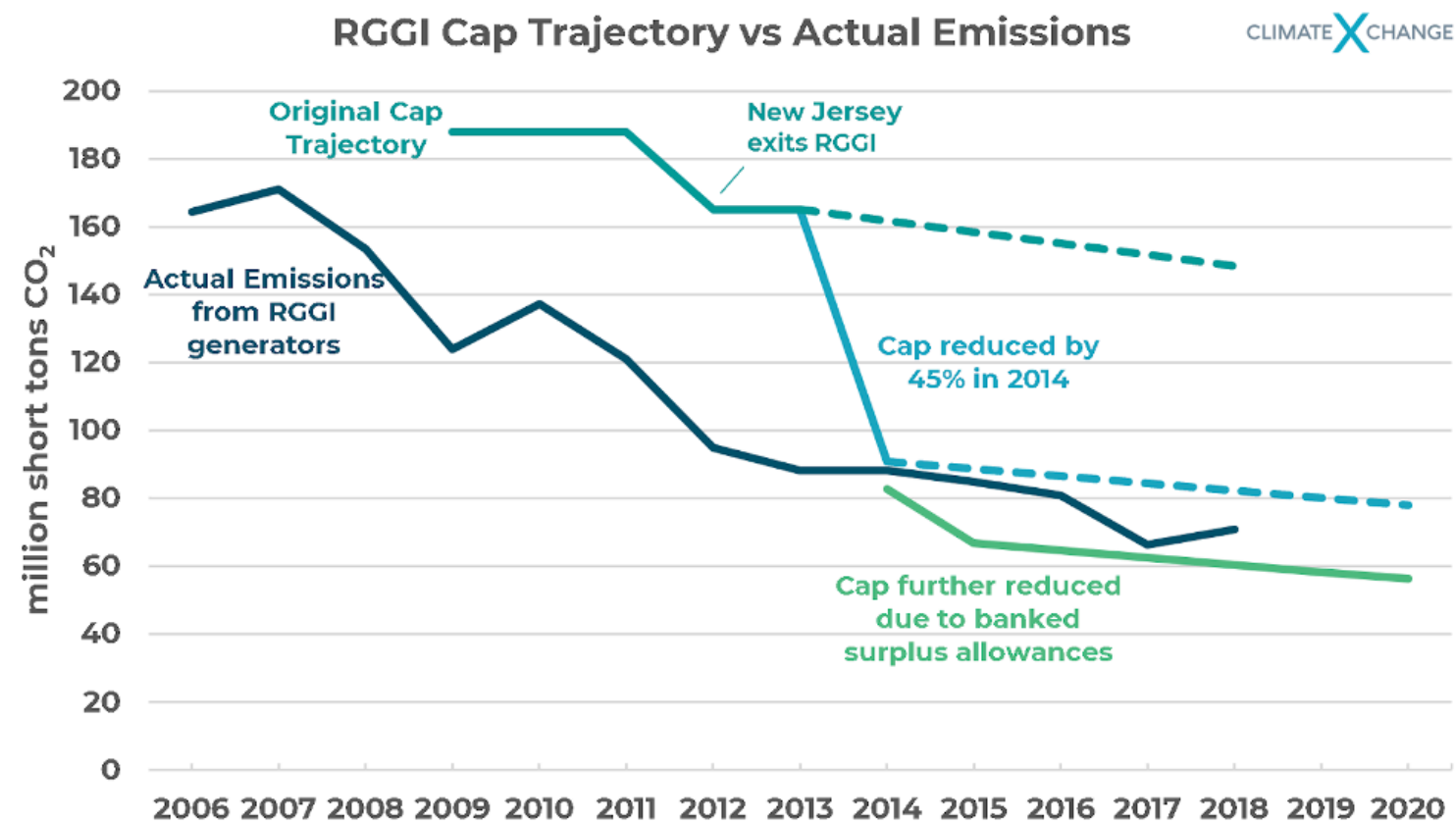
RGGI

Regional Greenhouse Gas Initiative (RGGI) allowance clearing price (Jan 2008-Dec 2019)
dollars per short ton of carbon dioxide (CO₂)



Source: U.S. Energy Information Administration, based on [Regional Greenhouse Gas Initiative](#)





AB32

CARBON PRICE

\$/Tonne CO₂e



EU-ETS

European carbon credits price

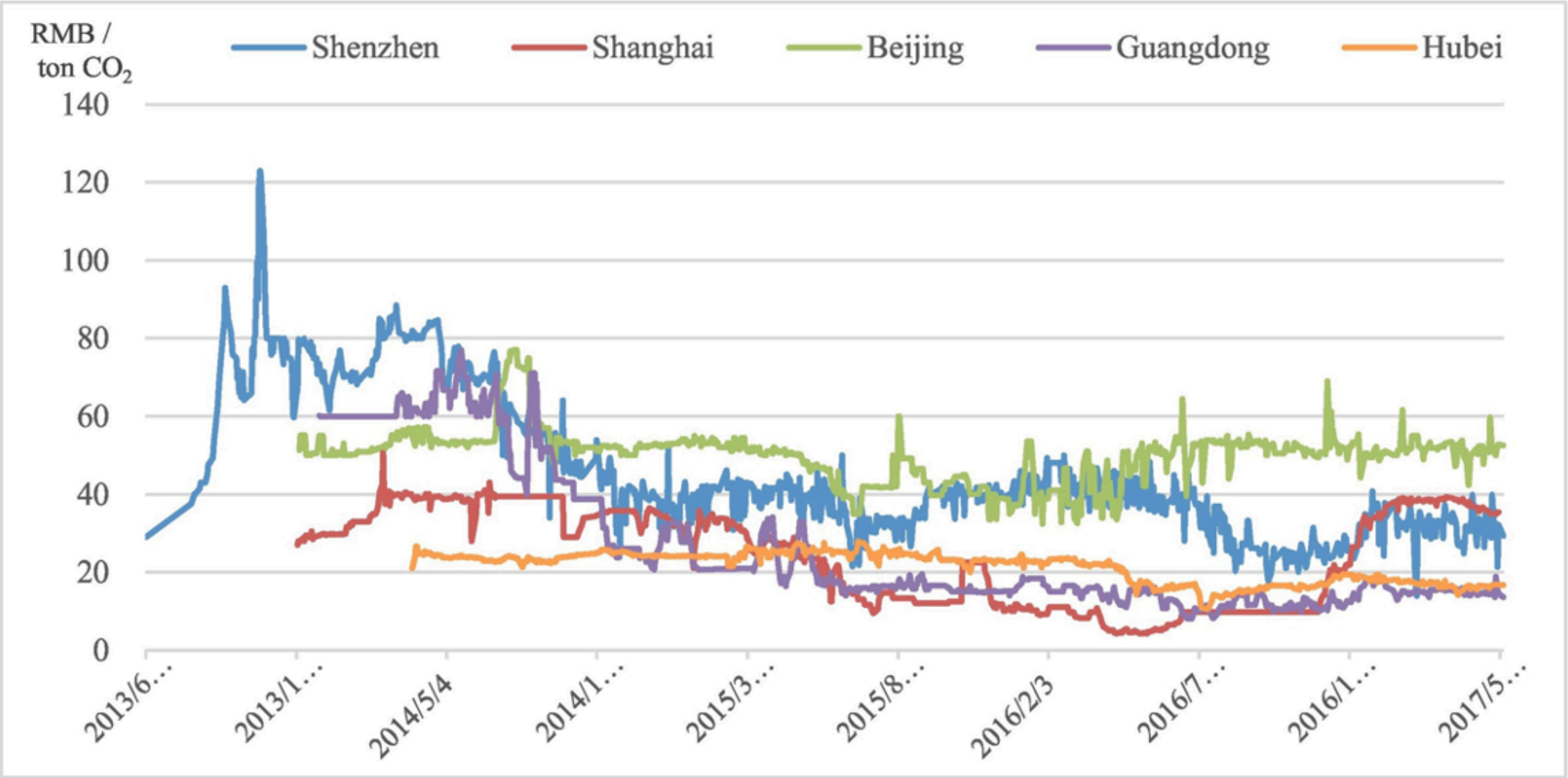
Euros per tonne



Source: Thomson Reuters

© FT

China ETS



Banking and borrowing in permit markets

Banking and borrowing in permits

In cap-and-trade, permits may be shifted across time if we allow for banking and borrowing:

Banking: save unused permits for future compliance

Borrowing: use future permits early, then make up the shortfall later

Let $\bar{E}_t$ be permits issued in period t , E_t emissions, and B_t the end-of-period bank:

$$B_t = B_{t-1} + \bar{E}_t - E_t$$

Banking only: $B_t \geq 0$

Borrowing allowed: B_t can be negative (typically with limits/penalties)

Banking and borrowing in permits

Two-period cost-minimization benchmark:

$$\min_{E_1, E_2} C_1(E_1) + C_2(E_2)$$

$$\text{subject to: } E_1 + E_2 = \bar{E}_1 + \bar{E}_2$$

$$-C'_1(E_1^*) = -C'_2(E_2^*)$$

In MAC terms:

$$MAC_1(E_1^*) = MAC_2(E_2^*)$$

A firm equalizes its MAC over time!

Banking and borrowing in permits

Intuition:

Standard cap-and-trade allows **within-period trade across firms**

Banking and borrowing add **across-period trade within a firm**

So a firm can effectively "trade with itself over time"

This shifts abatement away from high-MAC periods and toward low-MAC periods

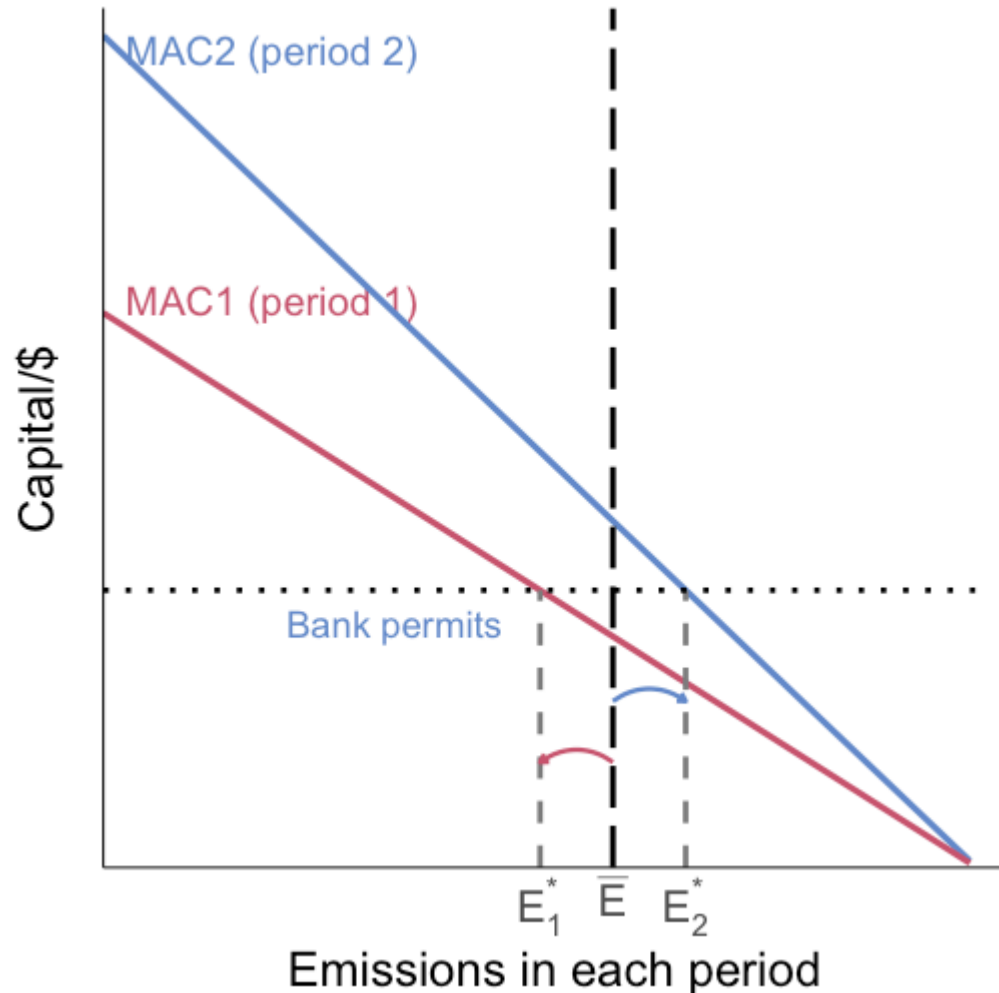
Banking and borrowing in permits

Why might a firm's MAC change over time?

1. Fuel and input prices move (e.g., coal vs natural gas prices)
2. Output demand changes (business cycle, weather-driven load)
3. Abatement equipment has temporary outages or maintenance
4. New technology adoption lowers future abatement cost
5. Learning-by-doing improves operational efficiency

With these shocks, banking/borrowing lowers compliance cost relative to a rigid period-by-period cap

Banking and borrowing in permits



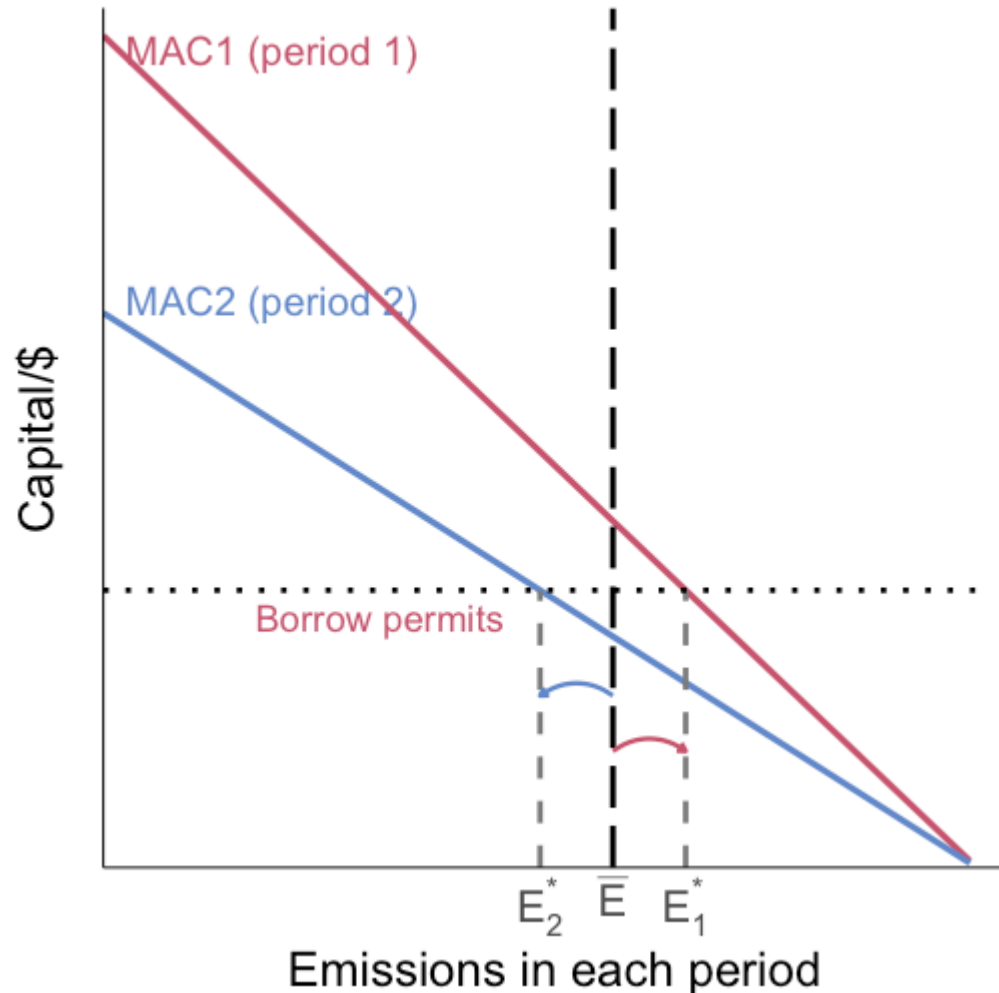
Decision rule at the no-flex caps:

If $MAC_2(\bar{E}_2) > MAC_1(\bar{E}_1)$, period 2 abatement is more expensive

Banking is optimal: $E_1^* < \bar{E}_1$ and $E_2^* > \bar{E}_2$

If $MAC_1(\bar{E}_1) > MAC_2(\bar{E}_2)$, borrowing is optimal

Banking and borrowing in permits



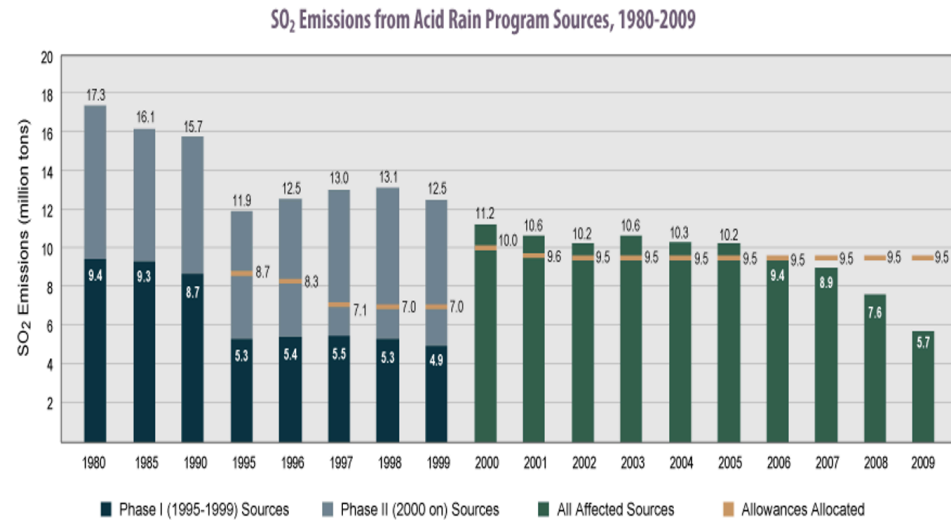
Borrowing case: period 1 starts with the higher MAC at cap

Cost minimization shifts emissions toward period 1 and away from period 2

$$E_1^* > \bar{E}_1 \text{ and } E_2^* < \bar{E}_2$$

Both banking and borrowing are intertemporal permit trade that aligns MACs across time

Banking and borrowing in practice: Acid Rain Program

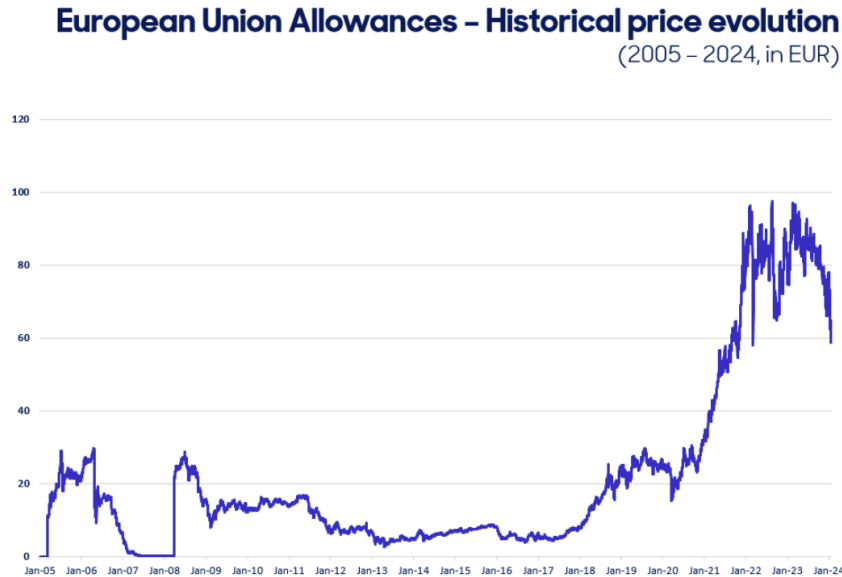


Source: EPA, 2010

U.S. SO₂ trading (Acid Rain Program) allowed banking across years

Firms over-complied in the 1990s and built an allowance bank before tighter caps

Banking and borrowing in practice: EU-ETS



Phase I (2005-2007): limited inter-phase banking; surplus allowances led to a price collapse near phase end

From Phase II onward, banking became a key margin for compliance planning

Large cumulative surplus carried forward across phases and affected prices

Banking and borrowing: design lessons

1. Banking is widely used because it lowers costs while preserving a cumulative cap
2. Borrowing is often constrained because of default, enforcement, and credibility concerns
3. Rules on banking/borrowing strongly affect allowance price dynamics
4. Without banking across phase boundaries, end-of-phase price volatility can be severe

The permit game: banking and borrowing

The permit game: decisions across rounds

We now allow permits to move between the two rounds

Bank: emit less than your permits cover in Round 1 and save the unused permits for Round 2

Borrow: emit more than your permits cover in Round 1 and repay the difference with Round 2 permits

Use the emissions choice in Round 1. Unrepaid borrowing pays a penalty at the end of Round 2

The permit game: predictions across rounds

How should expected Round 2 conditions change your Round 1 decisions?

- Bank when permits are more valuable in Round 2
- Borrow when permits are more valuable in Round 1, accounting for repayment
- Trading across rounds brings marginal abatement costs closer together

Compare these results with the game without banking and borrowing

The trading cost-reduction percentage is omitted here. Separate-round benchmarks do not measure cost-effectiveness across time

Rejoin the permit game: banking and borrowing

Scan to open the student login, then join with your team name



Or open aem4510.ivanrudik.com/permits

Debrief: banking and borrowing

Did teams bank before the tighter Round 2 cap? Did any teams borrow instead?

Compare your MAC across rounds: could shifting one more permit have reduced your total abatement cost?

Banking and borrowing extend the gains from trade across time

Comparison of standards, taxes, permits

What do we know so far

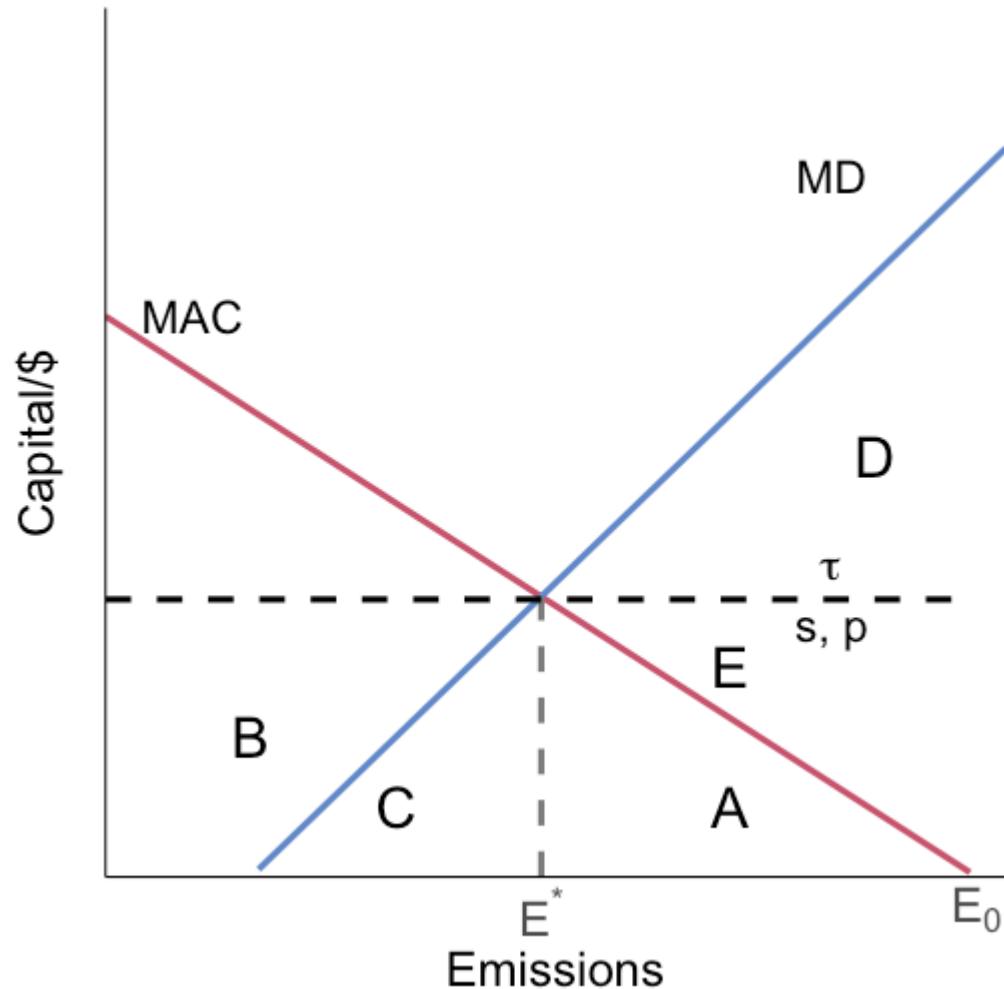
So far we have seen that:

1. Standards, taxes, and tradable permits can all achieve the efficient allocation
2. Taxes and tradable permits are cost-effective **no matter what**
 - (all firms set $MAC = \tau$ and $MAC = p$)

This still leaves a few questions to answer:

1. What are the equity effects?
2. What are the output effects?
3. What are the administrative burdens?
4. What are the **dynamic** incentives under these policies?

The equity set up

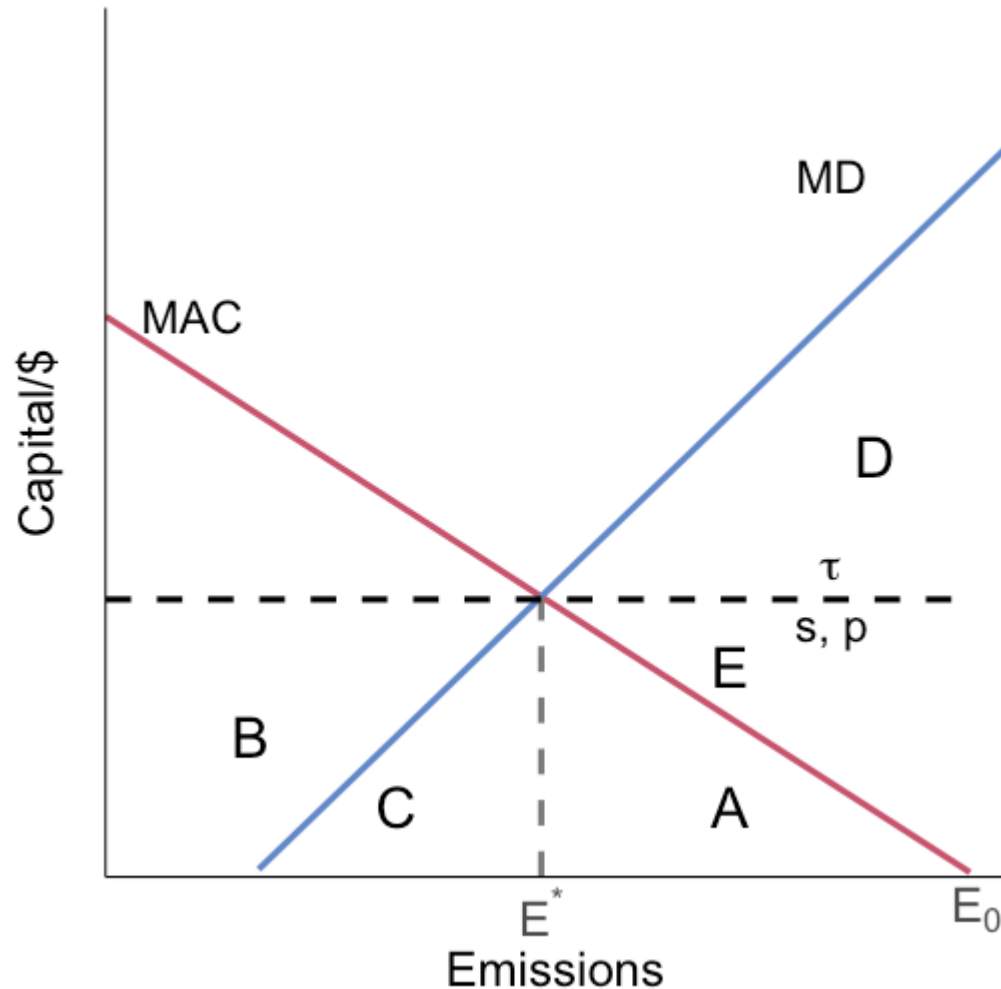


Lets consider this our base set up for 1 firm

The regulator can achieve E^* through:

- an emission standard of E^*
- a tax of τ
- an abatement subsidy of s
- "tradable permit" cap of E^* with auctioned permits at price p

The equity set up

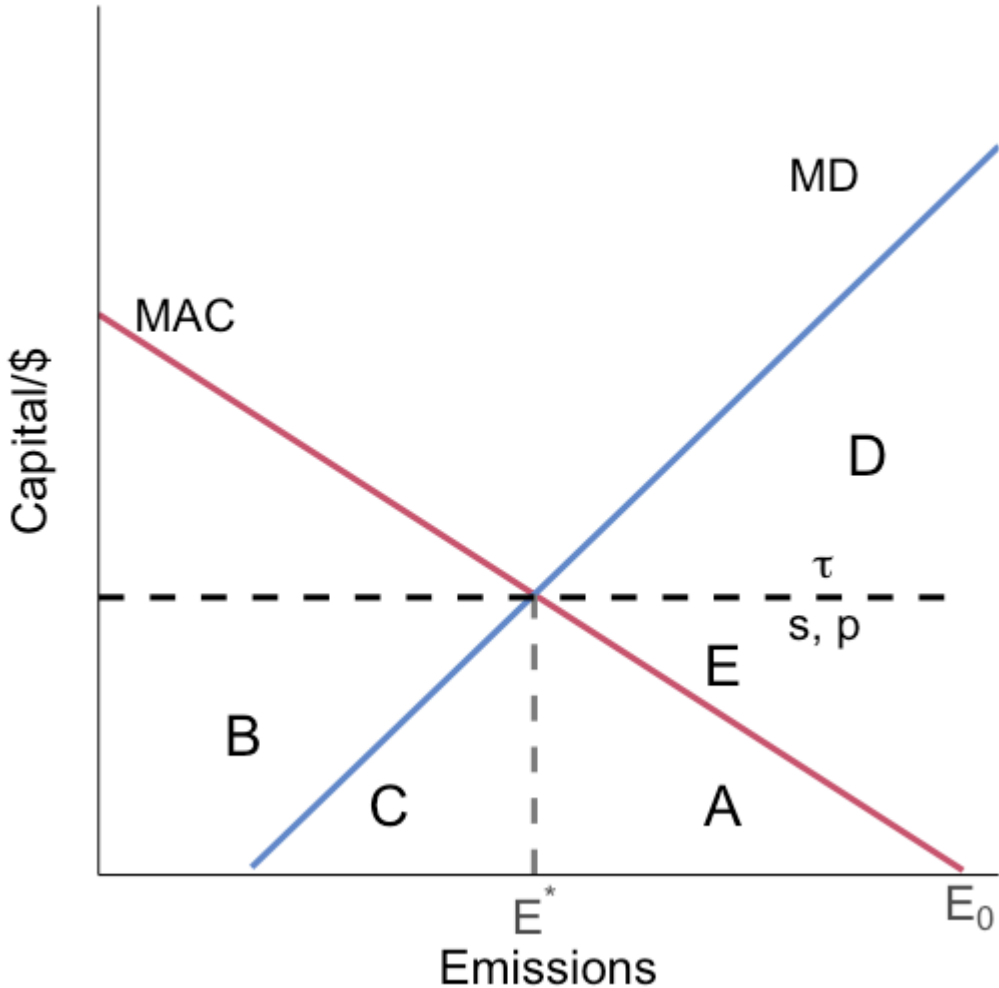


First let's look at **equity**

How do the costs and benefits of the policies fall on different groups?

From here on we will roll the tax and permit system into 1: they are actually identical in terms of their impacts if all permits are auctioned

The distributional outcomes



	Tax/Permits	Standard	Subsidy	Ranking
Firm	$-(A+B+C)$	$-A$	E	Sub > Std > Tax
Households	$A+D+E$	$A+D+E$	$A+D+E$	Indifferent
Government	$B+C$	0	$-(E+A)$	Tax > Std > Sub
Total	$D+E$	$D+E$	$D+E$	

The total welfare gain is the same for all policies

The difference is in the **distribution**

The standard strikes a middle ground out of the three

Output effects

So far we have assumed that actual firm output is not affected by abatement/emission decisions

This won't always be true in the real world

Key identity:

$$E = \left(\frac{E}{q} \right) \times q$$

Firms can reduce total emissions either by lowering emissions intensity per unit of output q : $\left(\frac{E}{q} \right)$, or by lowering output q

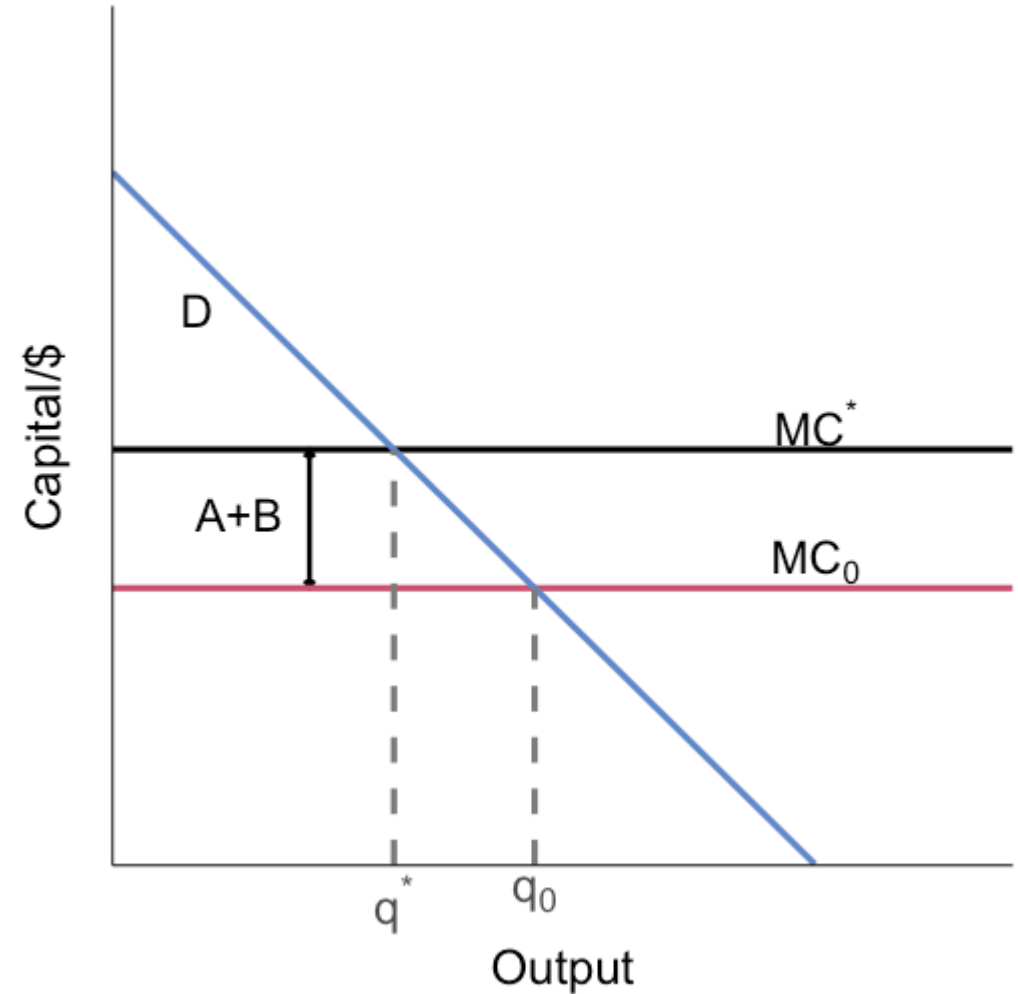
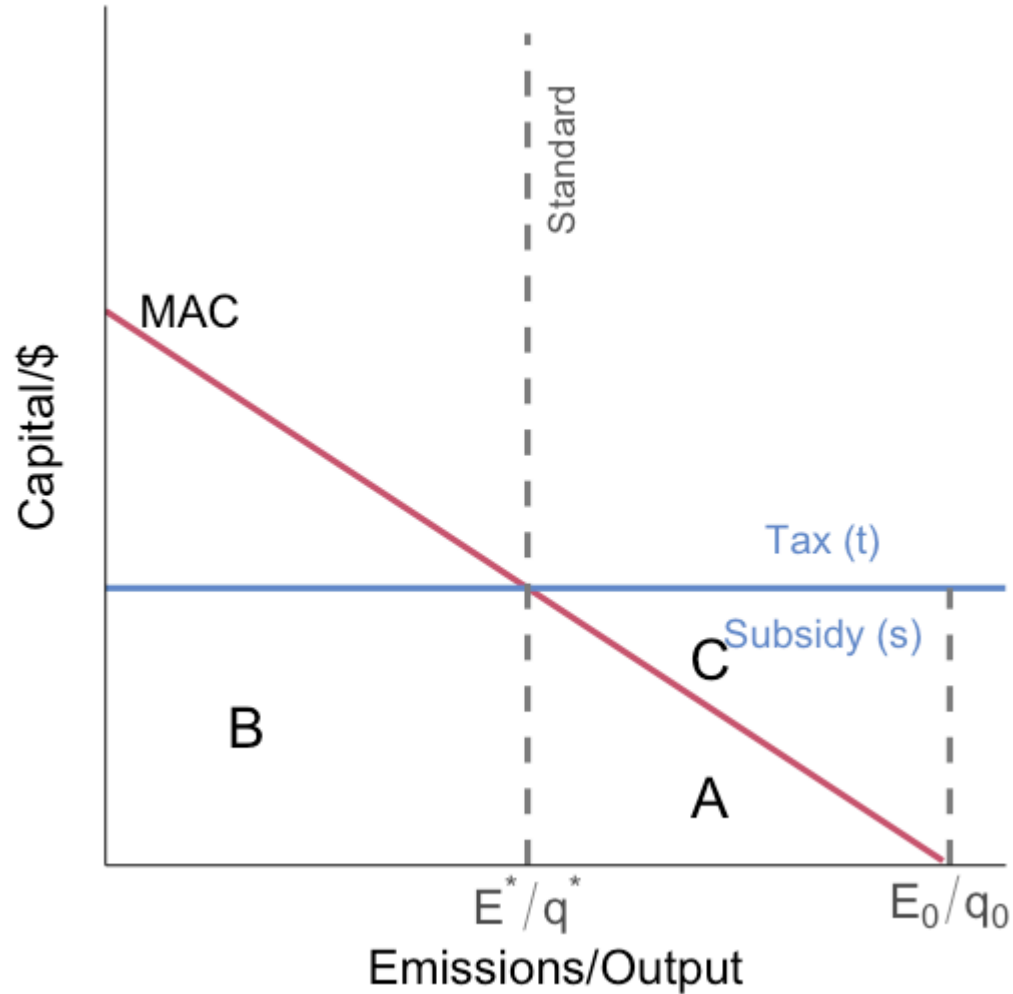
Output effects

Why can emissions policy reduce output?

1. Taxes and standards make cleaner production more costly per unit of output
2. That increases marginal cost of supplying output
3. With downward-sloping demand, higher MC implies lower profit-maximizing output

Even when the policy directly targets emissions, output can move as an indirect equilibrium response

The output set up



The output results

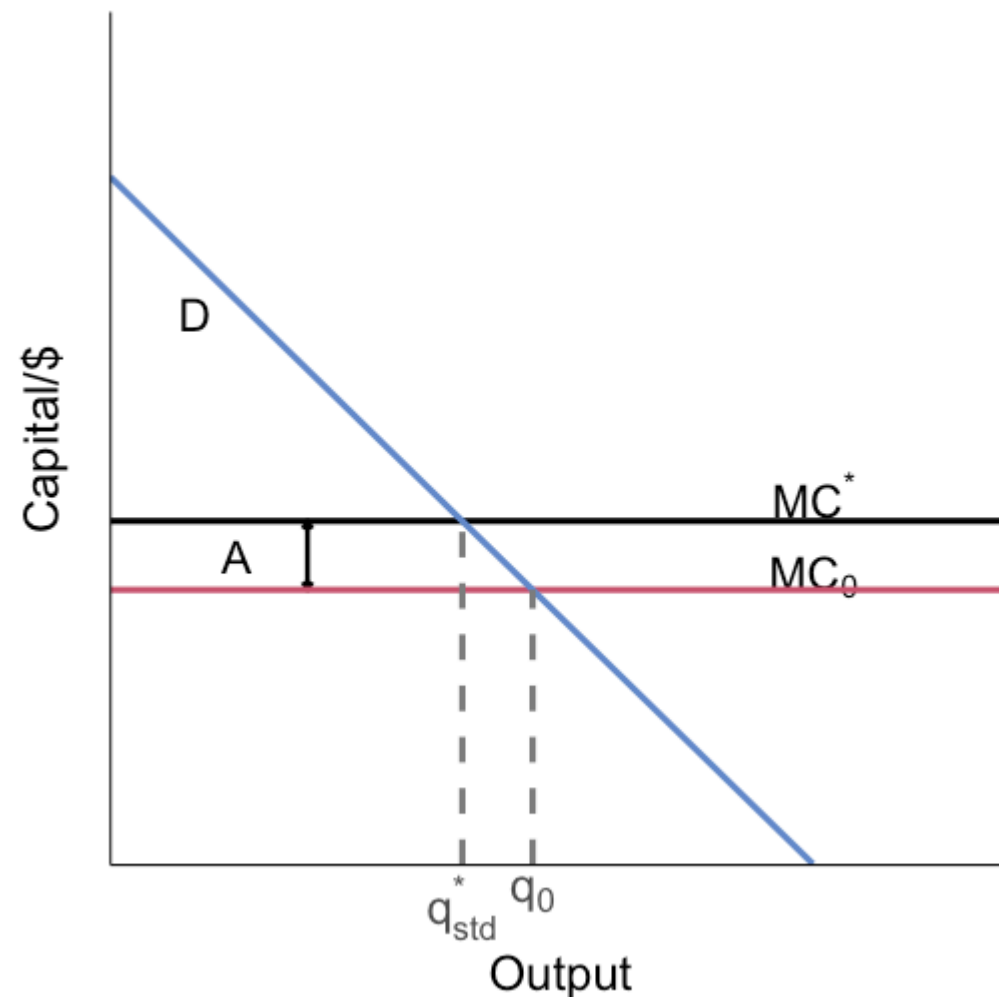
Emission tax:

- Firm chooses E^*/q^*
- Firm pays A+B in tax and abatement cost **per unit of output**
- This raises the MC of production by A+B to MC^*
- Output q^* falls
- Pollution $(E^*/q^*) * q^*$ falls even more since the tax lowers the optimal E^*/q^* , and increased MC lowers q^*

The output results

Emission standard:

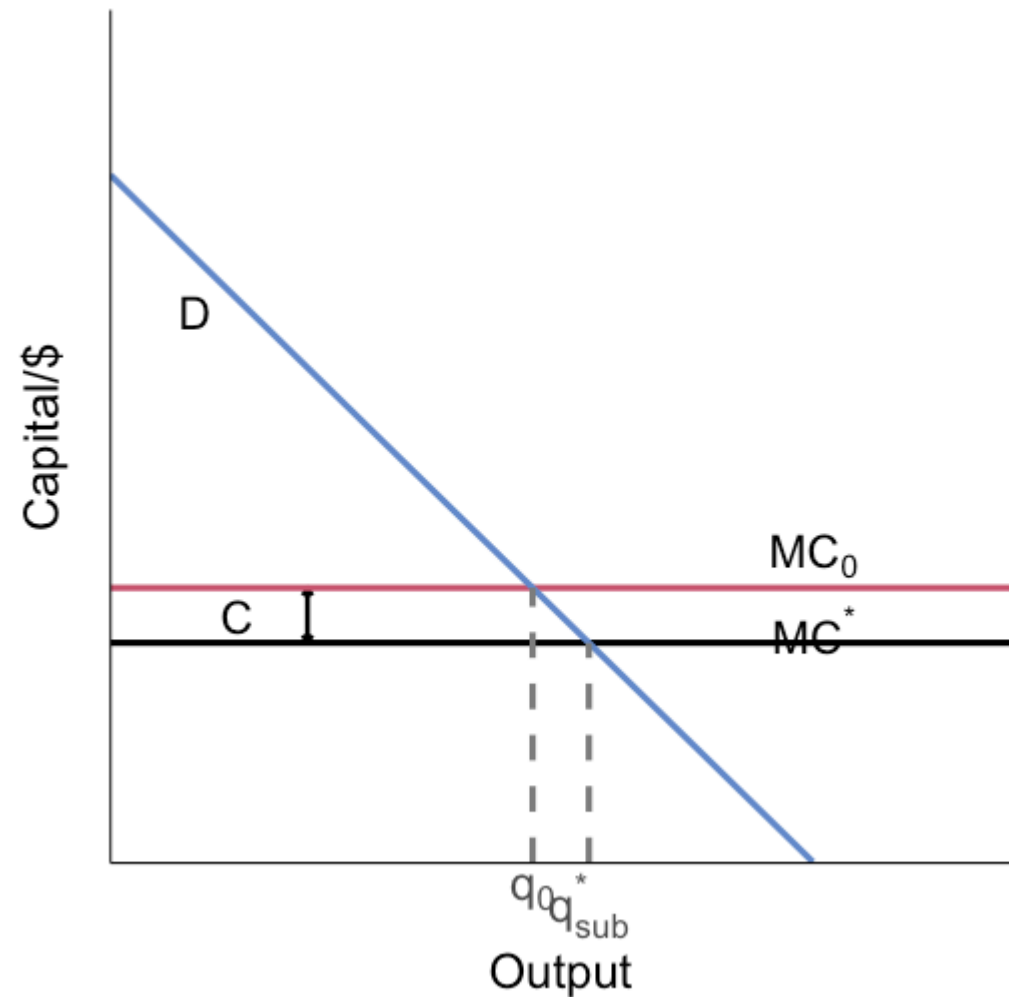
- Firm pays A in abatement cost per unit of output
- This raises the MC of production by A
- Output and $(E^*/q^*) * q^*$ fall, but not by as much as under the tax



The output results

Abatement subsidy:

- Reduces firm costs per unit of output by C
- This reduces the MC of production by C
- This **raises** output
- Even though E/q goes down because the subsidy induces a lower emission intensity, total emissions may go up because q will rise



Administration

These policies are not equivalent in their cost of implementation

Enforcing a technology standard is very easy: you can inspect to see if they have installed the tech or not

Emission standards, taxes, subsidies require monitoring of emissions

Monitoring systems cost money

Firms have incentives to try to cheat!

Administration

Pigouvian policies will not work as well for **non-point sources** like cars or farms

With non-point sources we observe total emissions but not who emitted it

With non-point sources it often makes sense to use technology standards

Point sources like power plants are much easier to handle with Pigouvian policies like taxes

Administration

Technology and emission standards typically guarantee some amount of emissions reductions

Taxes and subsidies guarantee firms pay a certain price but doesn't deliver us a guaranteed quantity

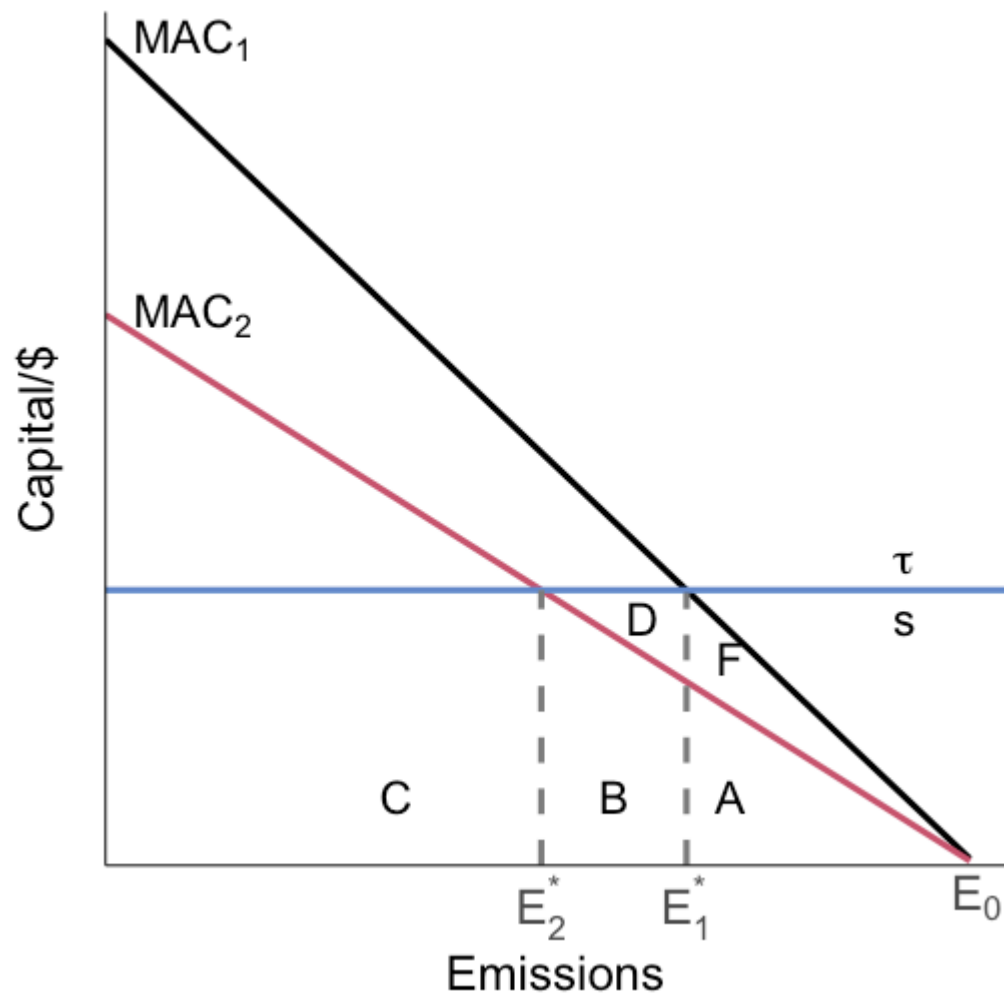
This might make things more politically difficult to pass

Administration

When does C&C / technology standards make sense?

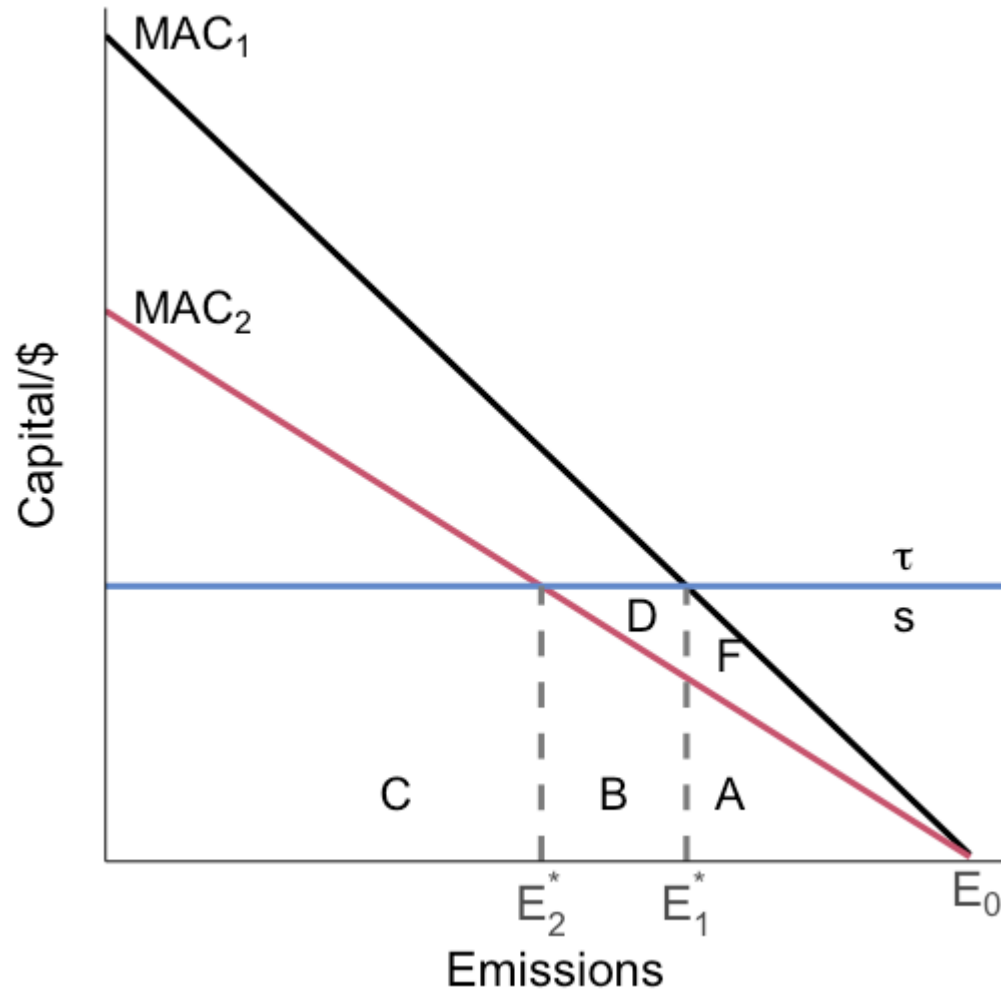
1. If there's a dominant technology where there's benefits to coordination or scale economies from production of the technology
2. High costs of monitoring/enforcement
3. High admin costs and little heterogeneity across firms

Dynamic incentives



What are the gains to the firm from moving from MAC_1 to MAC_2 ?

Dynamic incentives



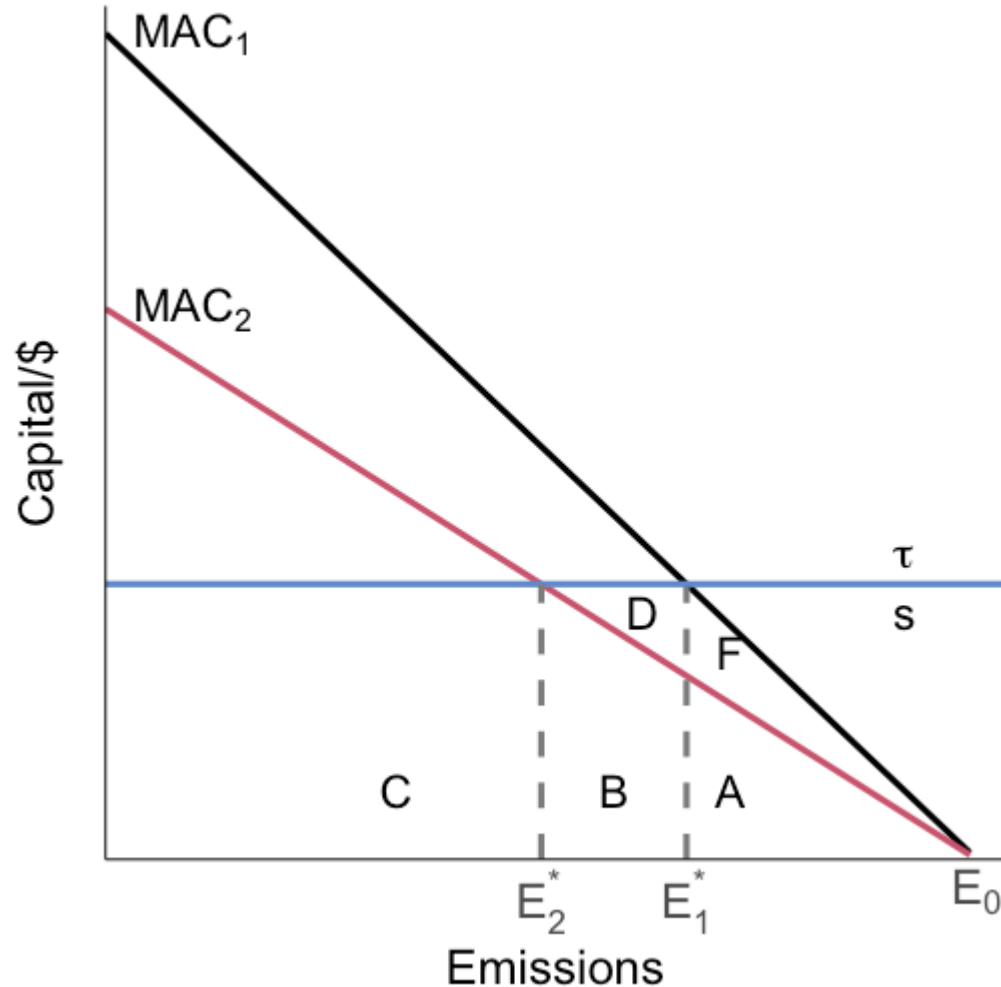
What are the gains to the firm from moving from MAC_1 to MAC_2 ?

Standard: F (abatement cost reduction)

Emission Tax: F + D (abatement cost and tax payment reduction)

Abatement Subsidy: F + D
(abatement cost reduction and abatement subsidy increase)

Dynamic incentives



What are the gains to the firm from moving to MAC_2 ?

Taxes and subsidies give greater incentives to innovate!

Once a firm meets a standard, there's no additional incentive beyond reducing abatement costs

Taxes and subsidies give the firm extra benefits for further reductions

How do carbon markets work?

